$$= \frac{2e^{2x} + 2e^{-2x}}{4} = \frac{e^{2x} + e^{-2x}}{2}$$

$\therefore \quad \cosh^2 x + \sinh^2 x = \cosh 2x$

Example 3: Determine whether the following functions are even or odd:

(a) $f(x) = 3x^4 - 2x^2 + 7$ (b) $f(x) = \frac{3x}{x^2+1}$ (c) $f(x) = \sin x + \cos x$

Solution:

(a) $f(-x) = 3(-x)^4 - 2(-x)^2 + 7 = 3x^4 - 2x^2 + 7 = f(x)$

Thus $f(x) = 3x^4 - 2x^2 + 7$ is even.

(b) $f(-x) = \frac{3(-x)}{(-x)^2+1} = -\left(\frac{3x}{x^2+1}\right) = -f(x)$

Thus $f(x) = \frac{3x}{x^2+1}$ is odd.

(c) $f(-x) = \sin(-x) + \cos(-x) = -\sin x + \cos x \neq \pm f(x)$

Thus $f(x) = \sin x + \cos x$ is neither even nor odd.

EXERCISE 1.1

1. Given that : (a) $f(x) = x^2 - x$ (b) $f(x) = \sqrt{x+4}$

Find (i) $f(-2)$ (ii) $f(0)$ (iii) $f(x-1)$ (iv) $f(x^2+4)$

2. Find $\frac{f(a+h) - f(a)}{h}$ and simplify where,

(i) $f(x) = 6x - 9$ (iii) $f(x) = \sin x$

(iv) $f(x) = x^3 + 2x^2 - 1$ (v) $f(x) = \cos x$

3. Express the following:

(a) The perimeter P of square as a function of its area A.

(b) The area A of a circle as a function of its circumference C.

(c) The volume V of a cube as a function of the area A of its base.

4. Find the domain and the range of the function g defined below, and sketch of graph of g:

(i) $g(x) = 2x - 5$ (ii) $g(x) = \sqrt{x^2 - 4}$

(iii) $g(x) = \sqrt{x+1}$ (iv) $g(x) = |x - 3|$

(v) $g(x)=\begin{cases}6x+7 & , \ x\le -2\\ 4-3 & , \ -2<x\end{cases}$ (vi) $g(x)=\begin{cases}x-1 & , \ x<3\\ 2x+1 & , \ 3\le x\end{cases}$

(vii) $g(x)=\dfrac{x^2-3x+2}{x+1}, \ x\neq -1$ (viii) $g(x)=\dfrac{x^2-16}{x-4}, \ x\neq 4$

5. Given $f(x)=x^3-ax^2+bx+1$

If $f(2)=-3$ and $f(-1)=0$. Find the values of a and b.

6. A stone falls from a height of 60m on the ground, the height h after x second is approximately given by $h(x)=40-10x^2$

(i) What is the height of the stone when:

(a) $x=1$ *sec*? (b) $x=1.5$ *sec*? (c) $x=1.7$ *sec*?

(ii) When does the stone strike the ground?

7. Show that the parametric equations:

(i) $x=at^2$, $y=2at$ represent the equation of parabola $y^2=4ax$

(ii) $x=a\cos\theta$, $y=b\sin\theta$ represent the equation of ellipse $\dfrac{x^2}{a^2}+\dfrac{y^2}{b^2}=1$

(iii) $x=a\sec$, $y=b\tan\theta$ represent the equation of hyperbola $\dfrac{x^2}{a^2}-\dfrac{y^2}{b^2}=1$

8. Prove the identities:

(i) $\sin h\,2x=2\sinh x\cosh x$ (ii) $\sec h^2x=1-\tan h^2x$

(iii) $\csc h^2x=\cot h^2x-1$

9. Determine whether the given function f is even or odd.

(i) $f(x)=x^3+x$ (ii) $f(x)=(x+2)^2$

(iii) $f(x)=x\sqrt{x^2+5}$ (iv) $f(x)=\dfrac{x-1}{x+1}, \ x\neq -1$

(v) $f(x)=x^{\frac{2}{3}}+6$ (vi) $f(x)=\dfrac{x^3-x}{x^2+1}$

1.3 COMPOSITION OF FUNCTIONS AND INVERSE OF A FUNCTION

Let f be a function from set X to set Y and g be a function from set Y to set Z. The composition of f and g is a function, denoted by gof , from X to Z and is defined by

$$(gof)(x)=g(f(x))=gf(x) \text{ , for all } x\in X$$

1.3.1 Composition of Functions

Explanation

Consider two real valued functions f and g defined by

$f(x)=2x+3$ and $g(x)=x^2$

Remember That:

Briefly we write gof as gf.

Verification:

$$f\left(f^{-1}(x)\right) = f\left(\frac{1}{2}(x-1)\right) = 2\left[\frac{1}{2}(x-1)\right]+1 = x$$

and $$f^{-1}\left(f(x)\right) = f^{-1}(2x+1) = \frac{1}{2}(2x+1-1) = x$$

Example 3: Without finding the inverse, state the domain and range of f^{-1}, where

$$f(x) = 2+\sqrt{x-1}$$

Solution: We see that f is not defined when $x < 1$.

$\therefore$ Domain $f = [1, +\infty)$

As x varies over the interval $[1, +\infty)$, the value of $\sqrt{x-1}$ varies over the interval $[0, +\infty)$. So the value of $f(x) = 2+\sqrt{x-1}$ varies over the interval $[2, +\infty)$.

Therefore range $f = [2, +\infty)$

By definition of inverse function f^{-1}, we have

domain f^{-1} = range $f = [2, +\infty)$

and range f^{-1} = domain $f = [1, +\infty)$

EXERCISE 1.2

1. The real valued functions f and g are defined below. Find

(a) $fog\,(x)$ (b) $gof(x)$ (c) $fof(x)$ (d) $gog(x)$

(i) $f(x) = 2x+1$; $g(x) = \frac{3}{x-1}$, $x \neq 1$

(ii) $f(x) = \sqrt{x+1}$; $g(x) = \frac{1}{x^2}, x \neq 0$

(iii) $f(x) = \frac{1}{\sqrt{x-1}}, x \neq 1$ $g(x) = (x^2+1)^2$

(iv) $f(x) = 3x^4 - 2x^2$; $g(x) = \frac{2}{\sqrt{x}}, x \neq 0$

2. For the real valued function, f defined below, find

(a) $f^{-1}(x)$ (b) $f^{-1}(-1)$ and verify $f(f^{-1}(x)) = f^{-1}(f(x)) = x$

(i) $f(x) = -2x+8$ (ii) $f(x) = 3x^3+7$

(iii) $f(x) = (-x+9)^3$ (iv) $f(x) = \frac{2x+1}{x-1}$, $x > 1$

3. Without finding the inverse, state the domain and range of f^{-1}.

(i) $f(x)=\sqrt{x+2}$ (iii) $f(x)=\dfrac{1}{x+3}$, $x\neq -3$

(ii) $f(x)=\dfrac{x-1}{x-4}$, $x\neq 4$ (iv) $f(x)=(x-5)^2$, $x\geq 5$

1.4 LIMIT OF A FUNCTION AND THEOREMS ON LIMITS

The concept of limit of a function is the basis on which the structure of calculus rests. Before the definition of the limit of a function, it is essential to have a clear understanding of the meaning of the following phrases:

1.4.1 Meaning of the Phrase "*x* approaches zero"

Suppose a variable x assumes in succession a series of values as

$$1, \frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}, \ldots \quad \text{i.e.,} \quad 1, \frac{1}{2}, \frac{1}{2^2}, \frac{1}{2^3}, \frac{1}{2^4}, \ldots, \frac{1}{2^n}, \ldots$$

We notice that x is becoming smaller and smaller as n increases and can be made as small as we please by taking n sufficiently large. This unending decrease of x is symbolically written as $x \to 0$ and is read as "x approaches zero" or "x tends to zero".

Note: The symbol $x \to 0$ is quite different from $x = 0$

(i) $x \to 0$ means that x is very close to zero but not actually zero.

(ii) $x = 0$ means that x is actually zero.

1.4.2 Meaning of the Phrase "*x* approaches infinity"

Suppose a variable x assumes in succession a series of values as

$$1, 10, 100, 1000, 10000, \ldots \quad \text{i.e.,} \quad 1, 10, 10^2, 10^3, \ldots, 10^n, \ldots$$

It is clear that x is becoming larger and larger as n increases and can be made as large as we please by taking n sufficiently large. This unending increase of x is symbolically written as "$x \to \infty$"and is read as "x approaches infinity" or "x tends to be infinity".

1.4.3 Meaning of the Phrase "*x* approaches *a*"

Symbolically it is written as "$x \to a$" which means that x is sufficiently close to but different from the number a, from both the left and right sides of a i.e; $x - a$ becomes smaller and smaller as we please but $x - a \neq 0$.

$$\therefore \lim_{\theta\to 0}\left(\frac{1-\cos\theta}{\theta}\right) = \lim_{\theta\to 0}\sin\theta \lim_{\theta\to 0}\frac{\sin\theta}{\theta} \lim_{\theta\to 0}\left(\frac{1}{1+\cos\theta}\right)$$

$$= (0)(1)\left(\frac{1}{1+1}\right)$$

$$= 0$$

EXERCISE 1.3

1. Evaluate each limit by using theorems of limits:

(i) $\lim_{x\to 3}(2x+4)$ (ii) $\lim_{x\to 1}(3x^2-2x+4)$ (iii) $\lim_{x\to 3}\sqrt{x^2+x+4}$

(iv) $\lim_{x\to 2} x\sqrt{x^2-4}$ (v) $\lim_{x\to 2}(\sqrt{x^3+1}-\sqrt{x^2+5})$ (vi) $\lim_{x\to -2}\frac{2x^3+5x}{3x-2}$

2. Evaluate each limit by using algebraic techniques.

(i) $\lim_{x\to -1}\frac{x^3-x}{x+1}$ (ii) $\lim_{x\to 0}\left(\frac{3x^3+4x}{x^2+x}\right)$ (iii) $\lim_{x\to 2}\frac{x^3-8}{x^2+x-6}$

(iv) $\lim_{x\to 1}\frac{x^3-3x^2+3x-1}{x^3-x}$ (v) $\lim_{x\to -1}\left(\frac{x^3+x^2}{x^2-1}\right)$ (vi) $\lim_{x\to 4}\frac{2x^2-32}{x^3-4x^2}$

(vii) $\lim_{x\to 2}\frac{\sqrt{x}-\sqrt{2}}{x-2}$ (viii) $\lim_{h\to 0}\frac{\sqrt{x+h}-\sqrt{x}}{h}$ (ix) $\lim_{x\to a}\frac{x^n-a^n}{x^m-a^m}$

3. Evaluate the following limits:

(i) $\lim_{x\to 0}\frac{\sin 7x}{x}$ (ii) $\lim_{x\to 0}\frac{\sin x^0}{x}$ (iii) $\lim_{\theta\to 0}\frac{1-\cos\theta}{\sin\theta}$

(iv) $\lim_{\theta\to \pi}\frac{\sin x}{\pi - x}$ (v) $\lim_{x\to 0}\frac{\sin ax}{\sin bx}$ (vi) $\lim_{x\to 0}\frac{x}{\tan x}$

(vii) $\lim_{x\to 0}\frac{1-\cos 2x}{x^2}$ (viii) $\lim_{x\to 0}\frac{1-\cos x}{\sin^2 x}$ (ix) $\lim_{\theta\to 0}\frac{\sin^2\theta}{\theta}$

(x) $\lim_{x\to 0}\frac{\sec x-\cos x}{x}$ (xi) $\lim_{\theta\to 0}\frac{1-\cos p\theta}{1-\cos q\theta}$ (xii) $\lim_{\theta\to 0}\frac{\tan\theta-\sin\theta}{\sin^3\theta}$

4. Express each limit in terms of *e*:

(i) $\lim_{n\to+\infty}\left(1+\frac{1}{n}\right)^{2n}$ (ii) $\lim_{n\to+\infty}\left(1+\frac{1}{n}\right)^{\frac{n}{2}}$ (iii) $\lim_{n\to+\infty}\left(1-\frac{1}{n}\right)^{n}$

Now $f(3) = 2(3)+1 = 7$

$\Rightarrow$ Condition (i) is satisfied.

$\lim\limits_{x\to 3^-} f(x) = \lim\limits_{x\to 3^-} (x-1) = 3-1 = 2$

$\lim\limits_{x\to 3^+} f(x) = \lim\limits_{x\to 3^+} (2x+1) = 6+1 = 7$

$\because$ $\lim\limits_{x\to 3^-} f(x) \neq \lim\limits_{x\to 3^+} f(x)$

i.e., condition (ii) is not satisfied

$\therefore \lim\limits_{x\to 3} f(x)$ does not exist

Hence $f(x)$ is not continuous at $x = 3$

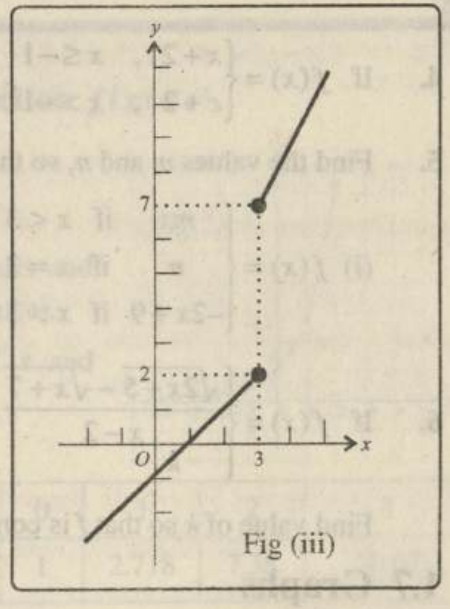


Fig (iii)

EXERCISE 1.4

1. Determine the left hand limit and the right hand limit and then, find limit of the following functions at $x \to c$:

(i) $f(x) = 2x^2 + x - 5,\ c = 1$ (ii) $f(x) = \dfrac{x^2-9}{x-3},\ c = -3$

(iii) $f(x) = |x-5|,\ c = 5$

2. Discuss the continuity of $f(x)$ at $x = c$:

(i) $f(x) = \begin{cases} 2x+5 & \text{if } x \le 2 \\ 4x+1 & \text{if } x > 2 \end{cases},\ c = 2$

(ii) $f(x) = \begin{cases} 3x-1 & \text{if } x < 1 \\ 4 & \text{if } x = 1 \\ 2x & \text{if } x > 1 \end{cases},\ c = 1$

3. If $f(x) = \begin{cases} 3x & \text{if } x \le -2 \\ x^2 - 1 & \text{if } -2 < x < 2 \\ 3 & \text{if } x \ge 2 \end{cases}$

Discuss continuity at $x = 2$ and $x = -2$.

4. If $f(x)=\begin{cases} x+2 & , \ x \le -1 \\ c+2 & , \ x > -1 \end{cases}$, find "$c$" so that $\lim_{x \to -1} f(x)$ exists.

5. Find the values m and n, so that given function f is continuous at $x = 3$.

(i) $f(x)=\begin{cases} mx & \text{if } x < 3 \\ n & \text{if } x = 3 \\ -2x+9 & \text{if } x > 3 \end{cases}$ (ii) $f(x)=\begin{cases} mx & \text{if } x < 3 \\ x^2 & \text{if } x \ge 3 \end{cases}$

6. If $f(x)=\begin{cases} \dfrac{\sqrt{2x+5}-\sqrt{x+7}}{x-2} & , \ x \ne 2 \\ k & , \ x = 2 \end{cases}$

Find value of k so that f is continuous at $x = 2$.

1.7 Graphs

We now learn the method to draw the graphs of the **Explicit Functions** like $y = f(x)$, where $f(x) = a^x$, e^x, $\log_a x$, and $\log_e x$.

1.7.1 Graph of the Exponential Function $f(x) = a^x$

Let us draw the graph of $y = 2^x$, here $a = 2$.

We prepare the following table for different values of x and $f(x)$ near the origin:

x	-4	-3	-2	-1	0	1	2	3	4
$y = f(x) = 2^x$	0.0625	0.125	0.25	0.5	1	2	4	8	16

Plotting the points (x, y) and joining them with smooth curve as shown in the figure, we get the graph of $y = 2^x$.

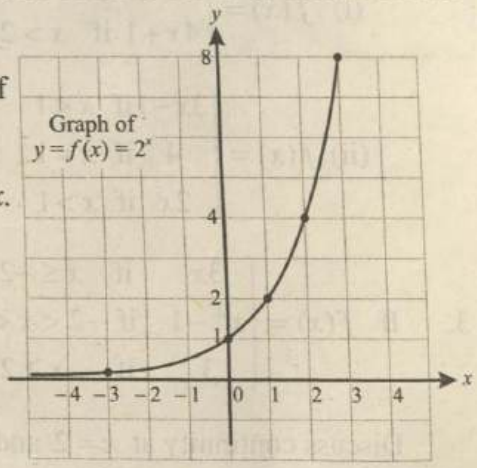
From the graph of 2^x, the characteristics of the graph of $y = a^x$ are observed as follows:

If $a > 1$, (i) a^x is always +ve for all real values of x.

(ii) a^x increases as x increases.

(iii) $a^x = 1$ when $x = 0$

(iv) $a^x \to 0$ as $x \to -\infty$



EXERCISE 1.5

1. Draw the graphs of the following equations:

(i) $x^2 + y^2 = 9$ (ii) $\frac{x^2}{16} + \frac{y^2}{4} = 1$

(iii) $y = e^{2x}$ (iv) $y = 3^x$

2. Graph the curves that has the parametric equations given below

(i) $x = t\ ,\ y = t^2\ ,\ -3 \le t \le 3$ where "t" is a parameter

(ii) $x = t - 1\ ,\ y = 2t - 1\ ,\ -1 < t < 5$ where "t" is a parameter

(iii) $x = sec\theta\ ,\ y = tan\theta$ where "θ" is a parameter

3. Draw the graphs of the functions defined below and find whether they are continuous.

(i) $y = \begin{cases} x-1 & \text{if } x < 3 \\ 2x+1 & \text{if } x \ge 3 \end{cases}$

(ii) $y = \frac{x^2 - 4}{x - 2}\ ,\ x \ne 2$

(iii) $y = \begin{cases} x+3 & \text{if } x \ne 3 \\ 2 & \text{if } x = 3 \end{cases}$

(iv) $y = \frac{x^2 - 16}{x - 4}\ ,\ x \ne 4$

4. Find the graphical solution of the following equations:

(i) $x = sin2x$

(ii) $\frac{x}{2} = cos\,x$

(iii) $2x = tan\,x$

Subtracting (i) from (ii), gives

$$\delta y = \frac{1}{(x+\delta x)^m} - \frac{1}{x^m} = \frac{x^m - (x+\delta x)^m}{x^m (x+\delta x)^m}.$$

$$= \frac{x^m - \left(x^m + mx^{m-1}\delta x + \frac{m(m-1)}{\underline{|2}} x^{m-2}(\delta x)^2 + \ldots + (\delta x)^m\right)}{x^m (x+\delta x)^m}$$

(expanding $(x+\delta x)^m$ by binomial theorem)

$$= \frac{-\delta x\left(mx^{m-1} + \frac{m(m-1)}{\underline{|2}} x^{m-2}\delta x + \ldots + (\delta x)^{m-1}\right)}{x^m . (x+\delta x)^m}$$

and $$\frac{\delta y}{\delta x} = \frac{-1}{x^m (x+\delta x)^m} \cdot \left(mx^{m-1} + \frac{m(m-1)}{\underline{|2}} x^{m-2} . \delta x + \ldots + (\delta x)^{m-1}\right)$$

Taking limit when $\delta x \to 0$, we get

$$\frac{dy}{dx} = \frac{-1}{x^m . x^m}(mx^{m-1}) \qquad \text{(all terms containing } \delta x, \text{ vanish)}$$

$$= (-m)\, x^{m-1} . x^{-2m} = (-m)\, x^{(-m)-1} = nx^{n-1} \qquad [\because -m = n]$$

or $$\frac{d}{dx}(x^n) = nx^{n-1}$$

So, far we have proved that $\frac{d}{dx}\left[x^n\right] = nx^{n-1}$, if $n \in Z$

The above rule holds if $n \in Q - Z$

For example $\frac{d}{dx}\left(x^{\frac{2}{3}}\right) = \frac{2}{3}x^{\frac{2}{3}-1} = \frac{2}{3x^{\frac{1}{3}}}$

The proof of $\frac{d}{dx}\left[x^n\right] = nx^{n-1}$ when $n \in Q - Z$ is left as an exercise.

Note that $\frac{d}{dx}\left[x^n\right] = nx^{n-1}$ is called power rule.

EXERCISE 2.1

1. Find by definition, the derivatives w.r.t 'x' of the following functions defined as:

(i) $2x^2 + 1$ (ii) $2 - \sqrt{x}$ (iii) $\frac{1}{\sqrt{x}}$ (iv) $\frac{1}{x^3}$ (v) $\frac{1}{x-a}$

(vi) $x(x-3)$ (vii) $\frac{2}{x^4}$ (viii) $(x+4)^{1/3}$ (ix) $x^{3/2}$ (x) $x^{5/2}$

(xi) $x^m, m \in N$ (xii) $\frac{1}{x^m}, m \in N$ (xiii) x^{40} (xiv) x^{-100}

2. Find $\frac{dy}{dx}$ from first principles if

(i) $\sqrt{x+2}$ (ii) $\frac{1}{\sqrt{x+a}}$

2.2.2 Differentiation of Expressions of the Types

$$(ax+b)^n \text{ and } \frac{1}{(ax+b)^n}, \quad n = 1, 2, 3, \cdots.$$

We find the derivatives of $(ax + b)^n$ and $\frac{1}{(ax+b)^n}$ from the first principles when $n \in N$.

Example 6: Find from definition the differential coefficient of $(ax + b)^n$ w.r.t. 'x' when n is a positive integer.

Solution. Let $y = (ax + b)^n$, (n is a positive integer)

Then $y + \delta y = [a(x + \delta x) + b]^n = [(ax + b) + a\,\delta x]^n$

Using the binomial theorem we have

$$y + \delta y = (ax+b)^n + \binom{n}{1}(ax+b)^{n-1}(a\,\delta x) + \binom{n}{2}(ax+b)^{n-2}(a\,\delta x)^2 + \cdots + (a\,\delta x)^n$$

$$\delta y = (y + \delta y) - y = \binom{n}{1}(ax+b)^{n-1}(a\,\delta x) + \binom{n}{2}(ax+b)^{n-2}.a^2(\delta x)^2 + \cdots + a^n(\delta x)^n$$

$$= \delta x\left[\binom{n}{1}(ax+b)^{n-1}.a + \binom{n}{2}(ax+b)^{n-2}.a^2\delta x + \cdots + a^n(\delta x)^{n-1}\right]$$

So, $$\frac{\delta y}{\delta x} = \binom{n}{1}(ax+b)^{n-1}a + \binom{n}{2}(ax+b)^{n-2}.a^2\delta x + \cdots + a^n(\delta x)^{n-1}$$

Taking limit when $\delta x \to 0$ we have

$$\lim_{\delta x \to 0}\frac{\delta y}{\delta x} = \lim_{\delta x \to 0}\left[\binom{n}{1}(ax+b)^{n-1}.a + \binom{n}{2}(ax+b)^{n-2}.a^2\,\delta x + \cdots + a^n(\delta x)^{n-1}\right]$$

or $$\frac{dy}{dx} = \binom{n}{1}(ax+b)^{n-1}.a$$ [All other terms tends to zero when $\delta x \to 0$]

Thus, $$\frac{d}{dx}(ax+b)^n = n(ax+b)^{n-1}.a$$

or $\frac{d}{dx}\left[\frac{1}{(ax+b)^n}\right] = \frac{-na}{(ax+b)^{n+1}} = n(ax+b)^{-(n+1)} \cdot a$

EXERCISE 2.2

1. Find from first principles, the derivatives of the following expressions w.r.t. their respective independent variables:

(i) $(ax+b)^3$ (ii) $(2x+3)^5$ (iii) $(3t+2)^{-2}$

(iv) $\frac{1}{(ax+b)^5}$ (v) $\frac{1}{(az-b)^7}$

2.3 THEOREMS ON DIFFERENTIATION

We have, so, far, proved the following two formula:

1. $\frac{d}{dx}(c) = 0$ **i.e., the derivative of a constant function is zero.**

2. $\frac{d}{dx}(x^n) = n x^{n-1}$ **power formula (or rule) when n is any rational number.**

Now we will prove other important formulas (or rules) which are used to determine derivatives of different functions efficiently. Henceforth, in all subsequent discussion, f, g, h etc. all denote functions differentiable at x, unless stated otherwise.

3. Derivative of $y = cf(x)$

Proof: Let $y = c\,f(x)$. Then

(i) $y + \delta y = c\,f(x+\delta x)$ and

(ii) $y + \delta y - y = c\,f(x+\delta x) - c\,f(x)$

or $\delta y = c\,[f(x+\delta x) - f(x)]$ (factoring out c)

(iii) $\frac{\delta y}{\delta x} = c\left(\frac{f(x+\delta x)-f(x)}{\delta x}\right)$

Taking limit when $\delta x \to 0$

(i) $y + \delta y = c\,f(x+\delta x)$ and

(ii) $y + \delta y - y = c\,f(x+\delta x) - c\,f(x)$

$\delta y = c[f(x+\delta x) - f(x)]$ (factoring out c)

(iii) $\frac{\delta y}{\delta x} = c\left(\frac{f(x+\delta x)-f(x)}{\delta x}\right)$

Taking limit when $\delta x \to 0$

(iv) $\underset{\delta x \to 0}{Lim}\ \frac{\delta y}{\delta x} = \underset{\delta x \to 0}{Lim}\left[c.\frac{f(x+\delta x)-f(x)}{\delta x}\right] = c\,.\ \underset{\delta x \to 0}{Lim}\ \frac{f(x+\delta x)-f(x)}{\delta x}$

A constant factor can be taken out from a limit sign.

Thus, $\frac{dy}{dx} = c\,f'(x)$, *that is*. $[c\,f(x)]' = c\,f'(x)$

$$= \frac{\sqrt{x}\left(1+\frac{1}{2}x^{-\frac{1}{2}}+0\right)-(x+\sqrt{x}+1).\left(\frac{1}{2}x^{-\frac{1}{2}}\right)}{x}$$

$$= \frac{\sqrt{x}\left(1+\frac{1}{2\sqrt{x}}\right)-(x+\sqrt{x}+1)\frac{1}{2\sqrt{x}}}{x}$$

$$= \frac{\sqrt{x}\left(\frac{2\sqrt{x}+1}{2\sqrt{x}}\right)-\frac{x+\sqrt{x}+1}{2\sqrt{x}}}{x} = \frac{2x+\sqrt{x}-x-\sqrt{x}-1}{\sqrt{x}.2\sqrt{x}} = \frac{x-1}{2x^{3/2}}$$

Example 8: Differentiate $\frac{2x^3-3x^2+5}{x^2+1}$ with respect to x.

Solution: Let $\phi(x) = \frac{2x^3-3x^2+5}{x^2+1}$. Then we take

$f(x) = 2x^3-3x^2+5$ and $g(x) = x^2+1$

Now $f'(x) = \frac{d}{dx}[2x^3-3x^2+5] = 2(3x^2)-3(2x)+0 = 6x^2-6x$

and $g'(x) = \frac{d}{dx}[x^2+1] = 2x+0 = 2x$

Using the quotient formula: $\phi'(x) = \frac{f'(x)\,g(x)-f(x)\,g'(x)}{[g(x)]^2}$, we obtain

$$\frac{d}{dx}\left[\frac{2x^3-3x^2+5}{x^2+1}\right] = \frac{(6x^2-6x)(x^2+1)-(2x^3-3x^2+5)(2x)}{(x^2+1)^2}$$

$$= \frac{6x^4-6x^3+6x^2-6x-(4x^4-6x^3+10x)}{(x^2+1)^2}$$

$$= \frac{6x^4-6x^3+6x^2-6x-4x^4+6x^3-10x}{(x^2+1)^2}$$

$$= \frac{2x^4+6x^2-16x}{(x^2+1)^2}$$

EXERCISE 2.3

Differentiate w.r.t. 'x'

1. $x^4+2x^3+x^2$ **2.** $x^{-3}+2x^{-\frac{3}{2}}+3$ **3.** $\frac{a+x}{a-x}$

4. $\frac{2x-3}{2x+1}$ 5. $(x-5)(3-x)$ 6. $\left(\sqrt{x}-\frac{1}{\sqrt{x}}\right)^2$

7. $\frac{(1+\sqrt{x})(x-x^{3/2})}{\sqrt{x}}$ 8. $\frac{(x^2+1)^2}{x^2-1}$ 9. $\frac{x^2+1}{x^2-3}$

10. $\frac{\sqrt{1+x}}{\sqrt{1-x}}$ 11. $\frac{2x-1}{\sqrt{x^2+1}}$ 12. $\sqrt{\frac{a-x}{a+x}}$

13. $\frac{\sqrt{x^2+1}}{\sqrt{x^2-1}}$ 14. $\frac{\sqrt{1+x}-\sqrt{1-x}}{\sqrt{1+x}+\sqrt{1-x}}$ 15. $\frac{x\sqrt{a+x}}{\sqrt{a-x}}$

16. If $y=\sqrt{x}-\frac{1}{\sqrt{x}}$, show that $2x\frac{dy}{dx}+y=2\sqrt{x}$

17. If $y=x^4+2x^2+2$, prove that $\frac{dy}{dx}=4x\sqrt{y-1}$

2.4 THE CHAIN RULE

The composition *fog* of functions f and g is the function whose values $f[g(x)]$ are found for each x in the domain of g for which $g(x)$ is in the domain of f. ($f[g(x)]$) is read as f of g of x).

Theorem: If g is differentiable at the point x and f is differentiable at the point $g(x)$ then the composition function *fog* is differentiable at the point x and $(fog)'(x)=f'[g(x)]\,.\,g'(x)$.

The proof of the claim rule is beyond the scope of this book.

If $y=(fog)(x)=f[g(x)]$, then

$$(fog)'(x)=f[g(x)]'=\frac{dy}{dx}$$

$$\Rightarrow \quad \frac{dy}{dx}=f'[g(x)]\,.\,g'(x) \qquad \text{(i)}$$

Let $u=g(x)$ (ii)

Then $y=f(u)$ (iii)

Differentiating (ii) and (iii) w.r.t x and u respectively, we have

$$\frac{du}{dx}=\frac{d}{dx}[g(x)]=g'(x)$$

and $$\frac{dy}{du}=\frac{d}{du}[f(u)]=f'(u)$$

Thus (i) can be written in the following forms

(a) $$\frac{d}{dx}(f(u))=f'(u)\frac{du}{dx}$$

(b) $\dfrac{dy}{dx} = \dfrac{dy}{du} \cdot \dfrac{du}{dx}$

The proof of the Chain rule is beyond the scope of this book.

Note: 1. Let $y = [g(x)]^n$ and $u = g(x)$

then $y = u^n$ and $\dfrac{dy}{du} = nu^{n-1}$ (power rule)

But $\dfrac{dy}{dx} = \dfrac{dy}{du} \cdot \dfrac{du}{dx} = nu^{n-1} \dfrac{du}{dx}$

or $\dfrac{d}{dx}[g(x)]^n = n[g(x)]^{n-1} \cdot g'(x)$ $\left(\because \dfrac{du}{dx} = g'(x)\right)$

2. Reciprocal rule can be written as

$$\frac{d}{dx}\left(\frac{1}{g(x)}\right) = \frac{d}{dx}[g(x)]^{-1} = -1 \cdot [g(x)]^{-1-1} \cdot g'(x)$$

$$= (-1)[g(x)]^{-2} \cdot g'(x)$$

Example 1: Find the derivative of $(x^3+1)^9$ with respect to x.

Solution: Let $y = (x^3+1)^9$ and $u = x^3 + 1$, then $y = u^9$

Now $\dfrac{du}{dx} = 3x^2$ and $\dfrac{dy}{du} = 9u^8$ (Power formula)

Using the formula $\dfrac{dy}{dx} = \dfrac{dy}{du} \cdot \dfrac{du}{dx}$, we have

$$\frac{dy}{dx} = 9u^8 \frac{du}{dx}$$

or $\dfrac{d}{dx}(x^3+1)^9 = 9(x^3+1)^8 (3x^2)$ $\left(\because u = x^3+1 \text{ and } \dfrac{du}{dx} = 3x^2\right)$

$$= 27x^2 (x^3+1)^8$$

Example 2. Differentiate $\sqrt{\dfrac{a-x}{a+x}}$, $(x \neq -a)$ with respect to x.

Solution: Let $y = \sqrt{\dfrac{a-x}{a+x}}$ and $u = \dfrac{a-x}{a+x}$ then $y = u^{\frac{1}{2}}$

Now $\dfrac{dy}{du} = \dfrac{1}{2} u^{\frac{1}{2}-1} = \dfrac{1}{2} u^{-\frac{1}{2}}$

and $\dfrac{du}{dx} = \dfrac{d}{dx}\left[\dfrac{a-x}{a+x}\right] = \dfrac{\left[\dfrac{d}{dx}(a-x)\right](a+x) - (a-x)\left[\dfrac{d}{dx}(a+x)\right]}{(a+x)^2}$

or $\frac{d}{dx}(y^3) - \frac{d}{dx}(2xy^2) + \frac{d}{dx}(x^2y) + \frac{d}{dx}(3x) = 0$

$$\frac{d}{dx}(y^3) - 2\left[1 \cdot y^2 + x\frac{d}{dx}(y^2)\right] + \left(2xy + x^2\frac{dy}{dx}\right) + 3 = 0$$

Using the chain rule on $\frac{d}{dx}(y^3)$ *and* $\frac{d}{dx}(y^2)$, *we have*

$$3y^2\frac{dy}{dx} - 2\left[y^2 + x\left(2y\frac{dy}{dx}\right)\right] + 2xy + x^2\frac{dy}{dx} + 3 = 0$$

or $(3y^2 - 4xy + x^2)\frac{dy}{dx} = 2y^2 - 2xy - 3$

$$\Rightarrow \quad \frac{dy}{dx} = \frac{2y^2 - 2xy - 3}{3y^2 - 4xy + x^2}$$

Example 5: Differentiate $x^2 + \frac{1}{x^2}$ w.r.t. $x - \frac{1}{x}$.

Solution: Let $y = x^2 + \frac{1}{x^2}$ and $u = x - \frac{1}{x}$. Then

$$\frac{dy}{dx} = 2x + (-2)\cdot\frac{1}{x^3} = 2\left(x - \frac{1}{x^3}\right) = \frac{2(x^4 - 1)}{x^3} = \frac{2(x^2 - 1)(x^2 + 1)}{x^3}$$

and $\frac{du}{dx} = 1 - (-1)\cdot\frac{1}{x^2} = 1 + \frac{1}{x^2} = \frac{x^2 + 1}{x^2}$

Thus, $\frac{dy}{du} = \frac{dy}{dx}\cdot\frac{dx}{du} = \frac{2(x^2 - 1)(x^2 + 1)}{x^3}\cdot\frac{x^2}{x^2 + 1} = \frac{2(x^2 - 1)}{x} = 2\left(x - \frac{1}{x}\right)$

EXERCISE 2.4

1. Find $\frac{dy}{dx}$ by making suitable substitutions in the following functions defined as:

(i) $y = \sqrt{\frac{1 - x}{1 + x}}$ (ii) $y = \sqrt{x + \sqrt{x}}$ (iii) $y = x\sqrt{\frac{a + x}{a - x}}$

(iv) $y = (3x^2 - 2x + 7)^6$ (v) $\sqrt{\frac{a^2 + x^2}{a^2 - x^2}}$

2. Find $\frac{dy}{dx}$ if:

(i) $3x + 4y + 7 = 0$ (ii) $xy + y^2 = 2$

(iii) $x^2 - 4xy - 5y = 0$ (iv) $4x^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$

(v) $x\sqrt{1 + y} + y\sqrt{1 + x} = 0$ (vi) $y(x^2 - 1) = x\sqrt{x^2 + 4}$

3. Find $\frac{dy}{dx}$ of the following parametric functions:

(i) $x = \theta + \frac{1}{\theta}$ *and* $y = \theta + 1$

(ii) $x = \dfrac{a(1+t^2)}{1+t^2}$, $y = \dfrac{2bt}{1+t^2}$

4. Prove that $y\dfrac{dy}{dx} + x = 0$ *if* $x = \dfrac{1-t^2}{1+t^2}$, $y = \dfrac{2t}{1+t^2}$

5. Differentiate

(i) $x^2 - \dfrac{1}{x^2}$ *w.r.t.* x^4 (ii) $(1+x^2)^n$ *w.r.t.* x^2 (iii) $\dfrac{x^2+1}{x^2-1}$ *w.r.t.* $\dfrac{x-1}{x+1}$

(iv) $\dfrac{ax+b}{cx+d}$ *w.r.t.* $\dfrac{ax^2+b}{ax^2+d}$ (v) $\dfrac{x^2+1}{x^2-1}$ *w.r.t.* x^3.

2.8 DERIVATIVES OF TRIGONOMETRIC FUNCTIONS

While finding derivatives of trigonometric functions, we assume that x is measured in radians. The limit theorems $\lim\limits_{x\to 0}\dfrac{\sin x}{x} = 1$ and $\lim\limits_{x\to 0}\dfrac{1-\cos x}{x} = 0$ are used to find the derivative formulas for *sin x* and *cos x*.

We prove from first principles that

$$\frac{d}{dx}(\sin x) = \cos x \text{ and } \frac{d}{dx}(\cos x) = -\sin x$$

Let $y = \sin x$. Then $y + \delta y = \sin(x + \delta x)$

and $\delta y = \sin(x+\delta x) - \sin x$

$$= 2\cos\left(\frac{x+\delta x+x}{2}\right)\sin\left(\frac{x+\delta x-x}{2}\right) = 2\cos\left(x+\frac{\delta x}{2}\right)\sin\left(\frac{\delta x}{2}\right)$$

$$\frac{\delta y}{\delta x} = \frac{2\cos\left(x+\frac{\delta x}{2}\right)\sin\left(\frac{\delta x}{2}\right)}{\delta x} = \cos\left(x+\frac{\delta x}{2}\right)\frac{\sin\left(\frac{\delta x}{2}\right)}{\frac{\delta x}{2}}$$

$$\lim_{\delta x\to 0}\frac{\delta y}{\delta x} = \lim_{\delta x\to 0}\left[\cos\left(x+\frac{\delta x}{2}\right)\frac{\sin\left(\frac{\delta x}{2}\right)}{\frac{\delta x}{2}}\right]$$

$$= \lim_{\frac{\delta x}{2}\to 0}\cos\left(x+\frac{\delta x}{2}\right)\lim_{\frac{\delta x}{2}\to 0}\frac{\sin\left(\frac{\delta x}{2}\right)}{\frac{\delta x}{2}} \quad \left(\because \frac{\delta x}{2}\to 0 \text{ when } \delta x\to 0\right)$$

EXERCISE 2.5

1. Differentiate the following trigonometric functions from the first principles.

(i) $\sin 2x$ (ii) $\tan 3x$ (iii) $\sin 2x + \cos 2x$ (iv) $\cos x^2$

(v) $\tan^2 x$ (vi) $\sqrt{\tan x}$ (vii) $\cos \sqrt{x}$

2. Differentiate the following w.r.t. the variable involved.

(i) $x^2 \sec 4x$ (ii) $\tan^3 \theta \sec^2 \theta$

(iii) $(\sin 2\theta - \cos 3\theta)^2$ (iv) $\cos \sqrt{x} + \sqrt{\sin x}$

3. Find $\frac{dy}{dx}$ *if*

(i) $y = x \cos y$ (ii) $x = y \sin y$

4. Find the derivative w.r.t. "x"

(i) $\cos \sqrt{\frac{1+x}{1+2x}}$ (ii) $\sin \sqrt{\frac{1+2x}{1+x}}$

5. Differentiate

(i) $\sin x$ w.r.t. $\cot x$ (ii) $\sin^2 x$ w.r.t. $\cos^4 x$

6. If $\tan y(1+\tan x) = 1 - \tan x$, show that $\frac{dy}{dx} = -1$

7. If $y = \sqrt{\tan x + \sqrt{\tan x + \sqrt{\tan x}}} + \cdots \infty$, prove that $(2y-1)\frac{dy}{dx} = \sec^2 x$.

8. If $x = a \cos^3 \theta$, $y = b \sin^3 \theta$, show that $a\frac{dy}{dx} + b \tan \theta = 0$

9. Find $\frac{dy}{dx}$ *if* $x = a(\cos t + \sin t)$, $y = a(\sin t - t \cos t)$

10. Differentiate w.r.t. "x"

(i) $\cos^{-1} \frac{x}{a}$ (ii) $\cot^{-1} \frac{x}{a}$ (iii) $\frac{1}{a} \sin^{-1} \frac{a}{x}$

(iv) $\sin^{-1} \sqrt{1-x^2}$ (v) $\sec^{-1}\left(\frac{x^2+1}{x^2-1}\right)$ (vi) $\cot^{-1}\left(\frac{2x}{1-x^2}\right)$

(vii) $\cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)$

11. Show that $\frac{dy}{dx} = \frac{y}{x}$ *if* $\frac{y}{x} = \tan^{-1} \frac{x}{y}$

12. If $y = \tan(p \tan^{-1} x)$, show that $(1+x^2)y_1 - p(1+y^2) = 0$.

Example 1: Find $\frac{dy}{dx}$ *if* $y = sinh^{-1}(ax + b)$

Solution: Let $u = ax + b$, then

$$y = sinh^{-1} u \quad \Rightarrow \quad \frac{dy}{du} = \frac{1}{\sqrt{1 + u^2}}$$

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx} = \frac{1}{\sqrt{1 + u^2}} \cdot \frac{du}{dx}$$

Thus $\frac{d}{dx}[sinh^{-1}(ax + b)] = \frac{1}{\sqrt{1 + (ax + b)^2}} \cdot a \quad \left(\because \frac{du}{dx} = \frac{d}{dx}(ax + b) = a\right)$

Example 2: Find $\frac{dy}{dx}$ if $y = cosh^{-1}(sec\, x) \quad , \quad 0 \le x \le \pi/2$

Solution: Let $u = sec\, x$, then

$$y = cosh^{-1} u \quad \Rightarrow \quad \frac{dy}{du} = \frac{1}{\sqrt{u^2 - 1}}$$

and $\frac{du}{dx} = \frac{d}{dx}(sec\, x) = sec\, x \tan x$

Thus $\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx} = \frac{1}{\sqrt{u^2 - 1}} \cdot \frac{du}{dx}$

$$= \frac{1}{\sqrt{sec^2 x - 1}}(sec\, x \tan x) = \frac{1}{\tan x}(sec\, x \tan x) = sec\, x$$

or $\frac{d}{dx}[cosh^{-1}(sec\, x)) = sec\, x.$

EXERCISE 2.6

1. Find $f'(x)$ if

(i) $f(x) = e^{\sqrt{x}-1}$ (ii) $f(x) = x^3 e^{\frac{1}{x}}, (x \neq 0)$ (iii) $f(x) = e^x(1 + ln\, x)$

(iv) $f(x) = \frac{e^x}{e^{-x} + 1}$ (v) $ln(e^x + e^{-x})$ (vi) $f(x) = \frac{e^{ax} - e^{-ax}}{e^{ax} + e^{-ax}}$

(vii) $f(x) = \sqrt{ln(e^{2x} + e^{-2x})}$ (viii) $ln(\sqrt{e^{2x} + e^{-2x}})$

2. Find $\frac{dy}{dx}$ if

(i) $y = x^2 ln\sqrt{x}$ (ii) $y = x\sqrt{ln\, x}$ (iii) $y = \frac{x}{ln\, x}$

(iv) $y = x^2 ln\frac{1}{x}$ (v) $y = ln\sqrt{\frac{x^2 - 1}{x^2 + 1}}$ (vi) $y = ln(x + \sqrt{x^2 + 1})$

(vii) $y = ln(9 - x^2)$ (viii) $y = e^{-2x} sin\, 2x$ (ix) $y = e^{-x}(x^3 + 2x^2 + 1)$

(x) $y = x\, e^{sin\, x}$ (xi) $y = 5e^{3x-4}$ (xii) $y = (x + 1)^x$

(xiii) $y = (\ln x)^{\ln x}$ (xiv) $y = \dfrac{\sqrt{x^2-1}\,(x+1)}{(x^3+1)^{3/2}}$

3. Find $\dfrac{dy}{dx}$ if

(i) $y = \cosh 2x$ (ii) $y = \sinh 3x$

(iii) $y = \tanh^{-1}(\sin x)$, $-\dfrac{\pi}{2} < x < \dfrac{\pi}{2}$ (iv) $y = \sinh^{-1}(x^3)$

(v) $y = (\ln \tanh x)$ (vi) $y = \sinh^{-1}\left(\dfrac{x}{2}\right)$

2.15 SUCCESSIVE DIFFERENTIATION (OR HIGHER DERIVATIVES)

Sometimes, it is useful to find the differential coefficient of a derived function. If we denote f' as the first derivative of f, then $(f')'$ is the derivative of f' and is called the second derivative of f. For convenience we write it as f''.

Similarly $(f'')'$, the derivative of f'', is called the third derivative of f and is written as f'''.

In general, for $n \geq 4$, the nth derivative of f is written as $f^{(n)}$.

Here we state different notations used for derivatives of higher orders.

1st derivative	2nd derivative	3rd derivative	nth derivative
y'	y''	y'''	$y^{(n)}$
$\dfrac{dy}{dx}$	$\dfrac{d^2y}{dx^2}$	$\dfrac{d^3y}{dx^3}$	$\dfrac{d^ny}{dx^n}$
y_1	y_2	y_3	y_n
D_y	D_y^2	D_y^3	D_y^n
$f'(x)$	$f''(x)$	$f'''(x)$	$f^{(n)}(x)$
$\dfrac{df}{dx}$	$\dfrac{d^2f}{dx^2}$	$\dfrac{d^3f}{dx^3}$	$\dfrac{d^nf}{dx^n}$

Example 1: Find higher derivatives of the polynomial

$$f(x) = \frac{1}{12}x^4 - \frac{1}{6}x^3 + \frac{1}{4}x^2 + 2x + 7$$

Solution: $f'(x) = \dfrac{1}{12}(4x^3) - \dfrac{1}{6}(3x^2) + \dfrac{1}{4}(2x) + 2 + 0 = \dfrac{1}{3}x^3 - \dfrac{1}{2}x^2 + \dfrac{1}{2}x + 2$

$$f''(x) = \frac{1}{3}(3x^2) - \frac{1}{2}(2x) + \frac{1}{2}(1) + 0 = x^2 - x + \frac{1}{2}$$

$$f'''(x) = 2x - 1$$

$$f^{iv}(x) = 2$$

All other higher derivatives are zero.

That is $\frac{dy}{dx} = -ay$ $(\because\ e^{-ax} = y)$

Now $\frac{d}{dx}\left[\frac{dy}{dx}\right] = \frac{d}{dx}[-ay] \Rightarrow \frac{d^2y}{dx^2} = -a\frac{dy}{dx} = (-a)(-ay)$ $\left(\because \frac{dy}{dx} = -ay\right)$

or $\frac{d^2y}{dx^2} = a^2y$ (i)

Differentiating (i) w.r.t. 'x', we get

$$\frac{d}{dx}\left[\frac{d^2y}{dx^2}\right] = \frac{d}{dx}[a^2y] \Rightarrow \frac{d^3y}{dx^3} = a^2\frac{dy}{dx} = a^2(-ay) = -a^3y$$

Thus $\frac{d^3y}{dx^3} + a^3y = 0.$

Example 7: If $y = sin^{-1}\frac{x}{a}$, then show that $y_2 = x\left(a^2 - x^2\right)^{-\frac{3}{2}}$.

Solution: $y = sin^{-1}\frac{x}{a}$, so

$$y_1 = \frac{dy}{dx} = \frac{d}{dx}\left[sin^{-1}\frac{x}{a}\right] = \frac{1}{\sqrt{1-\left(\frac{x}{a}\right)^2}} \times \frac{d}{dx}\left(\frac{x}{a}\right)$$

$$= \frac{1}{\sqrt{\frac{a^2-x^2}{a^2}}} \cdot \frac{1}{a} = \frac{a}{\sqrt{a^2-x^2}} \cdot \frac{1}{a} = (a^2-x^2)^{-1/2}$$

$$y_2 = \frac{d}{dx}\left[(a^2-x^2)^{-1/2}\right] = -\frac{1}{2}(a^2-x^2)^{-3/2} \times (-2x) = x(a^2-x^2)^{-3/2}$$

EXERCISE 2.7

1. Find y_2 if

(i) $y = 2x^5 - 3x^4 + 4x^3 + x - 2$ (ii) $y = (2x+5)^{3/2}$ (iii) $y = \sqrt{x} + \frac{1}{\sqrt{x}}$

2. Find y_2 if

(i) $y = x^2 . e^{-x}$ (ii) $y = ln\left(\frac{2x+3}{3x+2}\right)$

3. Find y_2 if

(i) $x^2 + y^2 = a^2$ (ii) $x^3 - y^3 = a^3$

(iii) $x = a\cos\theta,\ y = a\sin\theta$ (iv) $x = at^2,\ y = bt^4.$

(v) $x^2 + y^2 + 2gx + 2fy + c = 0$

4. Find y_4 if

(i) $y = sin\,3x$ (ii) $y = cos^3 x$ (iii) $ln\,(x^2 - 9)$

5. If $x = \sin\theta$, $y = \sin m\theta$, show that $(1-x^2)y_2 - xy_1 + m^2y = 0$

6. If $y = e^x \sin x$, show that $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + 2y = 0$

7. If $y = e^{ax} \sin bx$, show that $\frac{d^2y}{dx^2} - 2a\frac{dy}{dx} + (a^2+b^2)y = 0$

8. If $y = (\cos^{-1}x)^2$, prove that $(1-x^2)y_2 - xy_1 - 2 = 0$.

9. If $y = a\cos(\ln x) + b\sin(\ln x)$, prove that $x^2\frac{d^2y}{dx^2} + x\frac{dy}{dx} + y = 0$.

2.16 SERIES EXPANSIONS OF FUNCTIONS

A series of the form $a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 + \cdots + a_nx^n + \cdots$ is called a power series expansion of a function $f(x)$, where $a_0, a_1, a_2, \cdots, a_n, \cdots$ are constants and x is a variable.

We determine the coefficient $a_0, a_1, a_2, \cdots, a_n, \cdots$ to specify power series by finding successive derivatives of the power series and evaluating them at $x = 0$. That is,

$$f(x) = a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 + a_5x^5 + \cdots + a_nx^n + \cdots \quad f(0) = a_0$$

$$f'(x) = a_1 + 2a_2x + 3a_3x^2 + 4a_4x^3 + 5a_5x^4 + \cdots + na_nx^{n-1} + \cdots \quad f'(0) = a_1$$

$$f''(x) = 2a_2 + 6a_3x + 12a_4x^2 + 20a_5x^3 + \cdots + n(n-1)a_nx^{n-2} + \cdots \quad f''(0) = 2a_2$$

$$f'''(x) = 6a_3 + 24a_4x + 60a_5x^2 + \cdots\cdots \quad f'''(0) = 6a_3$$

$$f^{(4)}(x) = 24a_4 + 120a_5x \cdots\cdots \quad f^{(4)}(0) = 24a_4$$

So we have $a_0 = f(0)$, $a_1 = f'(0)$, $a_2 = \frac{f''(0)}{2!}$, $a_3 = \frac{f'''(0)}{3!}$, $a_4 = \frac{f^{(4)}(0)}{4!}$

Following the above pattern, we can write $a_n = \frac{f^{(n)}(0)}{n!}$

Thus substituting these values in the power series, we have

$$f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \frac{f^{(4)}(0)}{4!}x^4 + \cdots + \frac{f^{(n)}(0)}{n!}x^n + \cdots$$

This expansion of $f(x)$ is called the **Maclaurin series** expansion.

The above expansion is also named as **Maclaurin's Theorem** and can be stated as:

If $f(x)$ is expanded in ascending powers of x as an infinite series then,

$$f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \frac{f^{(4)}(0)}{4!}x^4 + \cdots + \frac{f^{(n)}(0)}{n!}x^n + \cdots$$

EXERCISE 2.8

1. Apply the Maclaurin series expansion to prove that:

(i) $ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + ...$

(ii) $cos\, x = 1 - \frac{x^2}{\lfloor 2} + \frac{x^4}{\lfloor 4} - \frac{x^6}{\lfloor 6} + ...$

(iii) $\sqrt{1+x} = 1 + \frac{x}{2} - \frac{x^2}{8} + \frac{x^3}{16} + ...$

(iv) $e^x = 1 + x + \frac{x^2}{\lfloor 2} + \frac{x^3}{\lfloor 3} + ...$

(v) $e^{2x} = 1 + 2x + \frac{4x^2}{\lfloor 2} + \frac{8x^3}{\lfloor 3} + ...$

2. Show that:

$cos\,(x+h) = cos\, x - h\, sin\, x - \frac{h^2}{\lfloor 2} cos\, x + \frac{h^3}{\lfloor 3} sin\, x + ...$

and evaluate $cos\ 61°$.

2. Show that $2^{x+h} = 2^x \{1 + (ln\ 2)h + \frac{(ln\ 2)^2 h^2}{\lfloor 2} + \frac{(ln\ 2)^3 h^3}{\lfloor 3} + ...\}$

2.18 GEOMETRICAL INTERPRETATION OF A DERIVATIVE

Let AB be the arc of the graph of f defined by the equation $y = f(x)$.

Let $P(x, f(x))$ and $Q(x+\delta x, f(x+\delta x))$ be two neighbouring points on the arc AB where x, $x+\delta x \in D_f$.

The line PQ is secant of the curve and it makes $\angle XSQ$ with the positive direction of the x-axis. (See the figure 2.21.1)

Drawing the ordinates PM, QN and perpendicular PR to NQ, we have

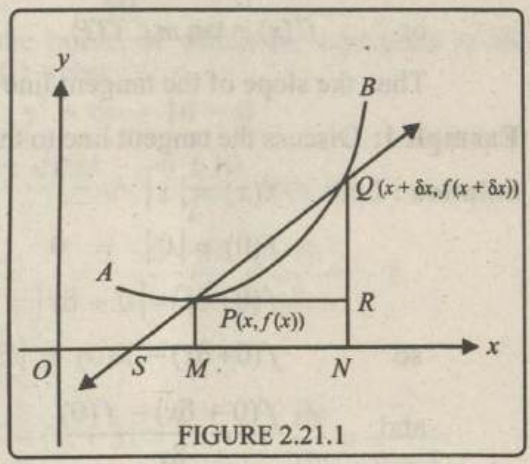


FIGURE 2.21.1

As $f''\left(\frac{\pi}{2}\right) = -\sin\frac{\pi}{2} - \sqrt{2}\cos\pi = -1 - \sqrt{2}\times(-1) = \sqrt{2} - 1 > 0$

and $f''\left(\frac{3\pi}{2}\right) = -\sin\frac{3\pi}{2} - \sqrt{2}\cos 3\pi = -1(-1) - \sqrt{2}(-1) = 1 + \sqrt{2} > 0$

Thus $f(x)$ has minimum values for $x = \frac{\pi}{2}$ and $x = \frac{3\pi}{2}$

As $f''\left(\frac{\pi}{4}\right) = -\sin\frac{\pi}{4} - \sqrt{2}\cos\frac{\pi}{2} = -\frac{1}{\sqrt{2}} - \sqrt{2}.0 = -\frac{1}{\sqrt{2}} < 0$

and $f''\left(\frac{3\pi}{4}\right) = -\sin\frac{3\pi}{4} - \sqrt{2}\cos\frac{3\pi}{2} = -\frac{1}{\sqrt{2}} - \sqrt{2}.0 = -\frac{1}{\sqrt{2}} < 0$

Thus $f(x)$ has minimum values for $x = \frac{\pi}{4}$ and $x = \frac{3\pi}{4}$

EXERCISE 2.9

1. Determine the intervals in which f is increasing or decreasing for the domain mentioned in each case.

(i) $f(x) = \sin x$: $x \in (-\pi, \pi)$

(ii). $f(x) = \cos x$: $x \in \left(\frac{-\pi}{2}, \frac{\pi}{2}\right)$

(iii) $f(x) = 4 - x^2$: $x \in (-2, 2)$

(iv) $f(x) = x^2 + 3x + 2$: $x \in (-4, 1)$

2. Find the extreme values for the following functions defined as:

(i) $f(x) = 1 - x^3$ (ii) $f(x) = x^2 - x - 2$

(iii) $f(x) = 5x^2 - 6x + 2$ (iv) $f(x) = 3x^2$

(v) $f(x) = 3x^2 - 4x + 5$ (vi) $f(x) = 2x^3 - 2x^2 - 36x + 3$

(vii) $f(x) = x^4 - 4x^2$ (viii) $f(x) = (x-2)^2(x-1)$

(ix) $f(x) = 5 + 3x - x^3$

3. Find the maximum and minimum values of the function defined by the following equation occurring in the interval $[0, 2\pi]$

$f(x) = \sin x + \cos x.$

4. Show that $y = \frac{\ln x}{x}$ has maximum value at $x = e$.

5. Show that $y = x^x$ has maximum value at $x = \frac{1}{e}$.

$\Rightarrow x = 1$ ($\because 2x^2 + 2x + 3 = 0$ gives complex roots)

l is positive for $1 - \varepsilon$ and $1 + \varepsilon$ where ε is very very small positive real number.

Also $2x^2 + 2x + 3 = 2(x^2 + x + \frac{1}{4}) + \frac{5}{2} = 2(x + \frac{1}{2})^2 + \frac{5}{2}$ is positive, for $x = 1 - \varepsilon$ and $x = 1 + \varepsilon$

The sign of $\frac{dl}{dx}$ depends on the factor $x - 1$

$x - 1$ is negative for $x = 1 - \varepsilon$ because $x - 1 = 1 - \varepsilon - 1 = -\varepsilon$ ..... (i)

$x - 1$ is positive for $x = 1 + \varepsilon$ because $x - 1 = 1 + \varepsilon - 1 = \varepsilon$ ..... (ii)

From (i) and (ii), we conclude that $\frac{dl}{dx}$ changes sign from $-$ve to $+$ve at $x = 1$

Thus l has a minimum value at $x = 1$

Putting $x = 1$ in $y = 4 - x^2$, we get the y-coordinate of the required point which is $4 - (1)^2 = 3$

Hence the required point on the curve is $(1, 3)$.

EXERCISE 2.10

1. Find two positive integers whose sum is 30 and their product will be maximum.

2. Divide 20 into two parts so that the sum of their squares will be minimum.

3. Find two positive integers whose sum is 12 and the product of one with the square of the other will be maximum.

4. The perimeter of a triangle is 16 centimetres. If one side is of length 6 cm, what are length of the other sides for maximum area of the triangle?

5. Find the dimensions of a rectangle of largest area having perimeter 120 centimetres.

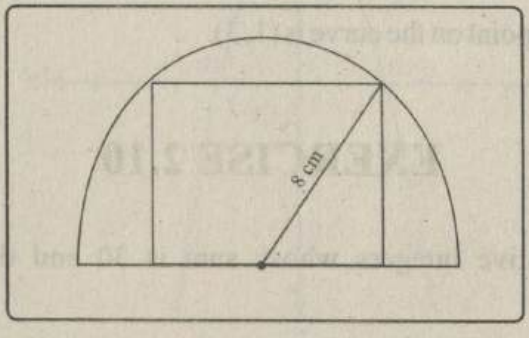
6. Find the lengths of the sides of a variable rectangle having area 36 cm^2 when its perimeter is minimum.

7. A box with a square base and open top is to have a volume of 4 cubic dm. Find the dimensions of the box which will require the least material.

8. Find the dimensions of a rectangular garden having perimeter 80 metres if its area is to be maximum.

9. An open tank of square base of side x and vertical sides is to be constructed to contain a given quantity of water. Find the depth in terms of x if the expense of lining the inside of the tank with lead will be least.

10. Find the dimensions of the rectangle of maximum area which fits inside the semi-circle of radius 8 cm as shown in the figure.



11. Find the point on the curve $y = x^2 - 1$ that is closest to the point $(3, -1)$.

12. Find the point on the curve $y = x^2 + 1$ that is closest to the point $(18, 1)$.

We take $x = 45° = \frac{\pi}{4}$ and $dx = 1° \approx 0.01745$

Hence $dy = \cos x\, dx \qquad \left(\because \frac{dy}{dx}(\sin x) = \cos x\right)$

$$\approx (\cos 45°)(0.01745) = \frac{1}{\sqrt{2}}(0.01745)$$

$$\approx 0.7071(0.01745)$$

$$\approx 0.01234$$

Using $f(x+\delta x) \approx f(x) + dy$ we have

$$\sin(46°) \approx \sin 45° + dy \approx 0.7071 + 0.01234 = 0.71944$$

$$= 0.7194$$

Using calculator, we find sin 46° approximately equal to 0.71934.

Example 6: The side of a cube is measured to be 20 cm with a maximum error of .12 cm in its measurement. Find the maximum error in the calculated volume of the cube.

Solution: Let x be the side and V be the volume of the cube, then

$$V = x^3 \text{ and } dV = (3x^2)\,dx$$

Taking $x = 20$ (cm) and $dx = 0.12$ (cm), we get

$$dV = [3(20)^2](0.12) = 1200 \times (0.12) = 144 \text{ (cubic cm)}$$

The error 144 cubic cm in volume calculation of a cube is either positive or negative.

EXERCISE 3.1

1. Find δy and dy in the following cases:

   (i) $y = x^2 - 1$ when x changes from 3 to 3.02

   (ii) $y = x^2 + 2x$ when x changes from 2 to 1.8

   (iii) $y = \sqrt{x}$ when x changes from 4 to 4.41

2. Using differentials find $\frac{dy}{dx}$ and $\frac{dx}{dy}$ in the following equations

   (i) $xy + x = 4$ (ii) $x^2 + 2y^2 = 16$ (iii) $x^4 + y^2 = xy^2$

   (iv) $xy - 1\, nx = c$

3. Use differentials to approximate the values of

   (i) $\sqrt[4]{17}$ (ii) $(31)^{1/5}$

   (iii) $\cos 29°$ (iv) $\sin 61°$

4. Find the approximate increase in the volume of a cube if the length of its each edge changes from 5 to 5.02..

$$= \frac{(x+1)^{\frac{3}{2}}}{\frac{3}{2}} + \frac{x^{\frac{3}{2}}}{\frac{3}{2}} + c = \frac{2}{3}(x+1)^{\frac{3}{2}} + \frac{2}{3}x^{\frac{3}{2}} + c$$

(vi) $\int \frac{\sin x + \cos^3 x}{\cos^2 x \sin x}\, dx$

Solution : $\int \frac{\sin x + \cos^3}{\cos^2 x \sin x}\, dx = \int \left(\frac{\sin x}{\cos^2 x \sin x} + \frac{\cos^3 x}{\cos^2 x \sin x} \right) dx$

$$= \int \left(\frac{1}{\cos^2} + \frac{\cos x}{\sin x} \right) dx$$

$$= \int \sec^2 x\, dx + \int \cot x\, dx$$

$$= \tan x + \ln|\sin x| + c$$

(vii) $\int \frac{3 - \cos 2x}{1 + \cos 2x}\, dx, \quad (\cos 2x \neq -1)$

Solution : $\int \frac{3 - \cos 2x}{1 + \cos 2x}\, dx = \int \frac{4 - (1 + \cos 2x)}{1 + \cos 2x}\, dx = \int \left(\frac{4}{1 + \cos 2x} - 1 \right) dx$

$$= \int \frac{4}{2\cos^2 2x}\, dx - \int 1\, dx = \int 2 \sec^2 x\, dx - \int 1\, dx$$

$$= 2 \tan x - x + c$$

EXERCISE 3.2

1. Evaluate the following indefi nite integrals

(i) $\int (3x^2 - 2x + 1)\, dx$ (ii) $\int \left(\sqrt{x} + \frac{1}{\sqrt{x}} \right) dx, \quad (x > 0)$

(iii) $\int x(\sqrt{x} + 1)\, dx, \quad (x > 0)$ (i v) $\int (2x + 3)^{\frac{1}{2}}\, dx$

(v) $\int (\sqrt{x} + 1)^2\, dx, \quad (x > 0)$ (vi) $\int \left(\sqrt{x} - \frac{1}{\sqrt{x}} \right)^2 dx, \quad (x > 0)$

(vii) $\int \frac{3x + 2}{\sqrt{x}}\, dx, \quad (x > 0)$ (viii) $\int \frac{\sqrt{y}(y+1)}{y}\, dy, \quad (y > 0)$

(ix) $\int \frac{(\sqrt{\theta} - 1)^2}{\sqrt{\theta}}\, d\theta \quad (\theta > 0)$ (x) $\int \frac{(1 - \sqrt{x})^2}{\sqrt{x}}\, dx, \quad (x > 0)$

(xi) $\int \frac{e^{2x} + e^x}{e^x}\, dx$

2. Evaluate

(i) $\int \frac{dx}{\sqrt{x+a}+\sqrt{x+b}}, \begin{pmatrix} x+a>0 \\ x+b>0 \end{pmatrix}$ (ii) $\int \frac{1-x^2}{1+x^2}\,dx,$

(iii) $\int \frac{dx}{\sqrt{x+a}+\sqrt{x}}, (x>0, a>0)$ (iv) $\int (a-2x)^{\frac{3}{2}}\,dx$

(v) $\int \frac{(1+e^x)^3}{e^x}\,dx$ (vi) $\int \sin\,(a+b)\,x\,dx$

(vii) $\int \sqrt{1-\cos\,2x}\,dx, (1-\cos\,2x>0)$ (viii) $\int (ln\,x)\times\frac{1}{x}\,dx,\ (x>0)$

(ix) $\int sin^2\,x\,dx$ (x) $\int \frac{1}{1+\cos\,x}\,dx, \left(-\frac{\pi}{2}<x<\frac{\pi}{2}\right)$

(xi) $\int \frac{ax+b}{ax^2+2bx+c}\,dx$ (xii) $\int cos\,3x\,sin\,2x\,dx$

(xiii) $\int \frac{cos\,2x-1}{1+cos\,2x}dx,\ (1+cos\,2x\neq 0)$ (xiv) $\int tan^2\,x\,dx$

3.3 INTEGRATION BY METHOD OF SUBSTITUTION

Sometimes it is possible to convert an integral into a standard form or to an easy integral by a suitable change of a variable. Now we evaluate $f(x)\,dx$ by the method of substitution. Let x be a function of a variable t, that is

if

$x = \phi(t)$, then $dx = \phi'(t)\,dt$

Putting $x = \phi(t)$, and $dx = \phi'(t)\,dt$, we have

$$\int f(x)dx = \int f(\phi(t))\,\phi'(t)\,dt$$

Now we explain the procedure with the help of some examples.

Example 1 : Evaluate $\int \frac{a\,dt}{2\sqrt{at+b}}, \quad (at+b>0)$

Solution: Let $at+b=u$. Then

$a\,dt = du$

Thus $\int \frac{a\,dt}{2\sqrt{at+b}} = \int \frac{du}{2\sqrt{u}} = \frac{1}{2}\int u^{\frac{-1}{2}}\,du$

$$= \frac{1}{2}\left(\frac{u^{-\frac{1}{2}+1}}{-\frac{1}{2}+1}\right)+c = \frac{1}{2}\left(\frac{u^{\frac{1}{2}}}{\frac{1}{2}}\right)+c \qquad = u^{\frac{1}{2}}+c = \sqrt{at+b}+c$$

Thus $\int \frac{dx}{\sqrt{(x+1)^2-1}} = \int \frac{\sec\theta \tan\theta\, d\theta}{\sqrt{\sec^2\theta - 1}} = \int \frac{\sec\theta \tan\theta\, d\theta}{\tan\theta} = \int \sec\theta\, d\theta$

$= \ln(\sec\theta + \tan\theta) + c \qquad = \ln(x+1+\sqrt{2x+x^2}) + c_1$

EXERCISE 3.3

Evaluate the following integrals:

1. $\int \frac{-2x}{\sqrt{4-x^2}}\, dx$
2. $\int \frac{dx}{x^2+4x+13}$
3. $\int \frac{x^2}{4+x^2}\, dx$
4. $\int \frac{1}{x \ln x}\, dx$
5. $\int \frac{e^x}{e^x+3}$
6. $\int \frac{x+b}{(x^2+2bx+c)^{\frac{1}{2}}}\, dx$
7. $\int \frac{\sec^2 x}{\sqrt{\tan x}}\, dx$
8. (a) Show that $\int \frac{dx}{\sqrt{x^2-a^2}} dx = \ln(x+\sqrt{x^2-a^2}) + c$

   (*b*) Show that $\int \sqrt{a^2-x^2}\, dx = \frac{a^2}{2} \sin^{-1}\frac{x}{a} + \frac{x}{2}\sqrt{a^2-x^2} + c$

Evaluate the following integrals:

9. $\int \frac{dx}{(1+x^2)^{\frac{3}{2}}}$
10. $\int \frac{1}{(1+x^2)\tan^{-1} x}\, dx$
11. $\int \sqrt{\frac{1+x}{1-x}}$
12. $\int \frac{\sin\theta}{1+\cos^2\theta}\, d\theta$
13. $\int \frac{ax}{\sqrt{a^2-x^4}}$
14. $\int \frac{dx}{\sqrt{7-6x-x^2}}$
15. $\int \frac{\cos x}{\sin x \ln \sin x}\, dx$
16. $\int \cos x \left(\frac{\ln \sin x}{\sin x}\right) dx$
17. $\int \frac{x\, dx}{4+2x+x^2}$
18. $\int \frac{x}{x^4+2x^2+5} dx$
19. $\int \left[\cos\left(\sqrt{x} - \frac{x}{2}\right)\right] \times \left(\frac{1}{\sqrt{x}} - 1\right) dx$
20. $\int \frac{x+2}{\sqrt{x+3}} dx$
21. $\int \frac{\sqrt{2}}{\sin x + \cos x}\, dx$
22. $\int \frac{dx}{\frac{1}{2}\sin x + \frac{\sqrt{3}}{2}\cos x}$

Example 11 : Show that $\int e^{ax}[a f(x)+f'(x)]\,dx = e^{ax} f(x)+c.$

Solution : $\int e^{ax}[af(x)+f'(x)]\,dx = \int e^{ax}\, a f(x)\,dx + \int e^{ax}.\, f'(x)\,dx$ (i)

In the second integral, let $\varphi(x)= e^{ax}$ *and* $g'(x) = f'(x)$,

then $\varphi'(x) = (e^{ax})\times a$ and $g(x) = f(x)$

so $\int e^{ax} f'(x)\,dx = e^{ax}\times f(x) - \int f(x)\times(ae^{ax})\,dx + c$

$= e^{ax} f(x) - \int a\, e^{ax} f(x)\,dx + c$

thus $\int e^{ax}[af(x)+f'(x)\,dx = \int a\, e^{ax} f(x)\,dx + \int e^{ax} f'(x)\,dx + c$

$= \int a\, e^{ax} f(x)\,dx + [e^{ax} f(x) - \int a e^{ax} f(x)\,dx + c]$

$= e^{ax} f(x) + c$

EXERCISE 3.4

Evaluate the following integrals by parts add a word representing all the functions are defined.

1. (i) $\int x \sin x\,dx$ (ii) $\int \ln x\,dx$ (iii) $\int x \ln x\,dx$

(iv) $\int x^2 \ln x\,dx$ (v) $\int x^3 \ln x\,dx$ (vi) $\int x^4 \ln x\,dx$

(vii) $\int \tan^{-1} x\,dx$ (viii) $\int x^2 \sin x\,dx$ (ix) $\int x^2 \tan^{-1} x\,dx$

(x) $\int x \tan^{-1} dx$ (xi) $\int x^3 \tan^{-1} x\,dx$ (xii) $\int x^3 \cos x\,dx$

(xiii) $\int \sin^{-1} x\,dx$ (xiv) $\int x \sin^{-1} x\,dx$ (xv) $\int e^x \sin x \cos x\,dx$

(xvi) $\int x \sin x \cos x\,dx$ (xvii) $\int x \cos^2 x\,dx$ (xviii) $\int x \sin^2 x\,dx$

(xix) $\int (\ln x)^2\,dx$ (xx) $\int (\ln(\tan x) \sec^2 x\,dx$ (xxi) $\int \frac{x \sin^{-1} x}{\sqrt{1-x^2}}\,dx$

2. **Evaluate the following integrals.**

(i) $\int \tan^4 x\,dx$ (ii) $\int \sec^4 x\,dx$ (iii) $\int e^x \sin 2x \cos x\,dx$

(iv) $\int \tan^3 x \sec x\,dx$ (v) $\int x^3 e^{5x}\,dx$ (vi) $\int e^{-x} \sin 2x\,dx$

(vii) $\int e^{2x} \cos 3x\,dx$ (viii) $\int \operatorname{cosec}^3 x\,dx$

3. Show that $\int e^{ax} \sin bx\,dx = \frac{1}{\sqrt{a^2+b^2}} e^{ax} \sin\left(bx - \tan^{-1}\frac{b}{a}\right) + c.$

4. **Evaluate the following indefinite integrals.**

(i) $\int \sqrt{a^2-x^2}\,dx$ (ii) $\int \sqrt{x^2-a^2}\,dx$ (iii) $\int \sqrt{4-5x^2}\,dx$

(iv) $\int \sqrt{3-4x^2}\,dx$ (v) $\int \sqrt{x^2+4}\,dx$ (vi) $\int x^2 e^{ax}\,dx$

5. Evaluate the following integrals.

(i) $\int e^x \left(\frac{1}{x} + \ln x\right) dx$ (ii) $\int e^x (\cos x + \sin x)\, dx$

(iii) $\int e^{ax} \left[a \sec^{-1} x + \frac{1}{x\sqrt{x^2-1}}\right] dx$ (iv) $\int e^{3x} \left(\frac{3 \sin x - \cos x}{\sin^2 x}\right) dx$

(v) $\int e^{2x} [-\sin x + 2 \cos x]\, dx$ (vi) $\int \frac{x e^x}{(1+x)^2}\, dx$

(vii) $\int e^{-x} (\cos x - \sin x)\, dx$ (viii) $\int \frac{e^{m \tan^{-1} x}}{(1+x^2)}\, dx$

(ix) $\int \frac{2x}{1 - \sin x}\, dx$ (x) $\int \frac{e^x (1+x)}{(2+x)^2}$

(xi) $\int \left(\frac{1 - \sin x}{1 - \cos x}\right) e^x\, dx$

3.6 INTEGRATION INVOLVING PARTIAL FRACTIONS

If $P(x)$, $Q(x)$ are polynomial functions and the denominator $Q(x)(\neq 0)$, in the rational function $\frac{P(x)}{Q(x)}$,can be factorized into linear and quadratic (irreducible) factors, then the rational function is written as a sum of simpler rational functions, each of which can be integrated by methods already known to us.

Here we will give examples of the following three cases when the denominator $Q(x)$ contains

Case I. Non-repeated linear factors.

Case II. Repeated and non-repeated linear factors.

Case III. Linear and non-repeated irreducible quadratic factors or non repeated irreducible quadratic factors.

EXAMPLES OF CASE I

Example 1: Evaluate $\int \frac{-x+6}{2x^2 - 7x + 6}\, dx, \quad (x > 2)$

Solution: The denominator $2x^2 - 7x + 6 = (x-2)(2x-3)$,

let $\frac{-x+6}{(x-2)(2x-3)} = \frac{A}{x-2} + \frac{B}{2x-3}$

EXERCISE 3.5

Evaluate the following integrals.

1. $\int \frac{3x+1}{x^2-x+6}\,dx$
2. $\int \frac{5x+8}{(x+3)(2x-1)}\,dx$
3. $\int \frac{x^2+3x-34}{x^2+2x-15}\,dx$
4. $\int \frac{(a-b)\,x}{(x-a)(x-b)}\,dx, (a>b)$
5. $\int \frac{3-x}{1-x+6x^2}\,dx$
6. $\int \frac{2x}{x^2-a^2}\,dx, \quad (x>a)$
7. $\int \frac{1}{6x^2+5x-4}\,dx$
8. $\int \frac{2x^3-3x^2-x-7}{2x^2-3x-2}\,dx$
9. $\int \frac{3x^2-12x+11}{(x-1)(x-2)(x-3)}\,dx$
10. $\int \frac{2x-1}{x\,(x-1)(x-3)}\,dx$
11. $\int \frac{5x^2+9x+6}{(x^2-1)(2x+3)}\,dx$
12. $\int \frac{4+7x}{(1+x)^2(2+3x)}\,dx$
13. $\int \frac{2x^2}{(x-1)^2(x+1)}\,dx$
14. $\int \frac{1}{(x-1)(x+1)^2}\,dx$
15. $\int \frac{x+4}{x^2-3x^2+4}\,dx$
16. $\int \frac{x^3-6x^2+25}{(x+1)^2(x-2)^2}\,dx$
17. $\int \frac{x^3+22x^2+14x-17}{(x-3)(x+2)^3}\,dx$
18. $\int \frac{x-2}{(x+1)(x^2+1)}\,dx$
19. $\int \frac{x}{(x-1)(x^2+1)}\,dx$
20. $\int \frac{9x-7}{(x+3)(x^2+1)}\,dx$
21. $\int \frac{1+4x}{(x-3)(x^2+4)}\,dx$
22. $\int \frac{12}{x^3+8}\,dx$
23. $\int \frac{9x+6}{x^3-8}\,dx$
24. $\int \frac{2x^2+5x+3}{(x-1)^2(x^2+4)}\,dx$
25. $\int \frac{2x^2-x-7}{(x+2)^2(x^2+x+1)}\,dx$
26. $\int \frac{3x+1}{(4x^2+1)(x^2-x+1)}\,dx$
27. $\int \frac{4x+1}{(x^2+4)(x^2+4x+5)}\,dx$
28. $\int \frac{6a^2}{(x^2+a^2)(x^2+4a^2)}\,dx$
29. $\int \frac{2x^2-2}{x^4+x^2+1}\,dx$
30. $\int \frac{3x-8}{(x^2-x+2)(x^2+x+2)}\,dx$
31. $\int \frac{3x^3+4x^2+9x+5}{(x^2+x+1)(x^2+2x+3)}\,dx$

EXERCISE 3.6

Evaluate the following definite integrals:

1. $\int_1^2 (x^2+1)\,dx$
2. $\int_{-1}^1 (x^{\frac{1}{3}}+1)\,dx$
3. $\int_{-2}^0 \frac{1}{(2x-1)^2}\,dx$
4. $\int_{-6}^2 \sqrt{3-x}\,dx$
5. $\int_1^{\sqrt{5}} \sqrt{(2t-1)^3}\,dt$
6. $\int_2^{\sqrt{5}} x\sqrt{x^2-1}\,dx$
7. $\int_1^2 \frac{x}{x^2+2}\,dx$
8. $\int_2^3 \left(x-\frac{1}{x}\right)^2 dx$
9. $\int_{-1}^1 \left(x+\frac{1}{2}\right)\sqrt{x^2+x+1}\,dx$
10. $\int_0^3 \frac{dx}{x^2+9}$
11. $\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \cos t\,dt$
12. $\int_1^2 \left(x+\frac{1}{x}\right)^{\frac{1}{2}}\left(1-\frac{1}{x^2}\right)dx$
13. $\int_1^2 \ln x\,dx$
14. $\int_0^2 \left(e^{\frac{x}{2}}-e^{-\frac{x}{2}}\right)dx$
15. $\int_0^{\frac{\pi}{4}} \frac{\cos\theta+\sin\theta}{2\cos^2\theta}\,d\theta$
16. $\int_0^{\frac{\pi}{6}} \cos^3\theta\,d\theta$
17. $\int_{\frac{\pi}{6}}^{\frac{\pi}{4}} \cos^2\theta\cot^2\theta\,d\theta$
18. $\int_0^{\frac{\pi}{4}} \cos^4 t\,dt$
19. $\int_0^{\frac{\pi}{3}} \cos^2\sin\theta\,d\theta$
20. $\int_0^{\frac{\pi}{4}} (1+\cos^2\theta)\tan^2\theta\,d\theta$
21. $\int_0^{\frac{\pi}{4}} \frac{\sec\theta}{\sin\theta+\cos\theta}\,d\theta$
22. $\int_{-1}^5 |x-3|\,dx$
23. $\int_{\frac{1}{8}}^1 \frac{\left(x^{\frac{1}{3}}+2\right)^2}{x^{\frac{2}{3}}}\,dx$
24. $\int_1^3 \frac{x^2-2}{x+1}\,dx$
25. $\int_2^3 \frac{3x^2-2x+1}{(x-1)(x^2+1)}\,dx$
26. $\int_0^{\frac{\pi}{4}} \frac{\sin x-1}{\cos^2 x}\,dx$
27. $\int_0^{\frac{\pi}{4}} \frac{1}{1+\sin x}\,dx$

28. $\int_0^1 \frac{3x}{\sqrt{4-3x}}\,dx$

29. $\int_{\frac{\pi}{6}}^{\frac{\pi}{2}} \frac{\cos x}{\sin x(2+\sin x)}\,dx$

30. $\int_0^{\frac{\pi}{2}} \frac{\sin x}{(1+\cos x)(2+\cos x)}\,dx$

3.8 APPLICATION OF DEFINITE INTEGRALS

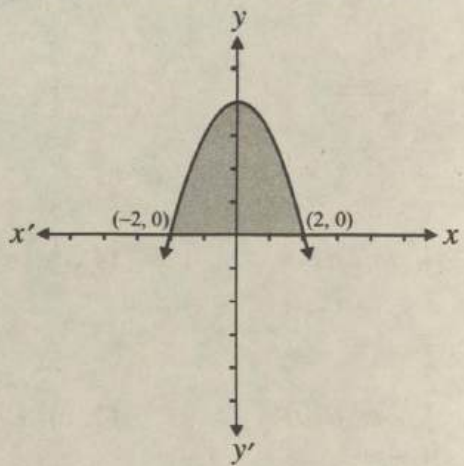
Here we shall give some examples involving area bounded by the curve and the x-axis.

Example 1: Find the area bounded by the curve $y = 4 - x^2$ and the x-axis.

Solution: We first find the points where the curve cuts the x-axis. Putting $y = 0$, we have

$$4 - x^2 = 0 \Rightarrow x = \pm 2.$$

So the curve cuts the x-axis at (-2,0) and (2,0)



The area above the x-axis and under the curve $y = 4 - x^2$ is shown in the figure as shaded region.

Thus the required area $= \int_{-2}^{2} (4 - x^2)\,dx = \left[4x - \frac{x^3}{3}\right]_{-2}^{2}$

$$= \left(4(2) - \frac{(2)^3}{3}\right) - \left(4(-2) - \frac{(-2)^3}{3}\right)$$

$$= \left(8 - \frac{8}{3}\right) - \left(-8 + \frac{8}{3}\right)$$

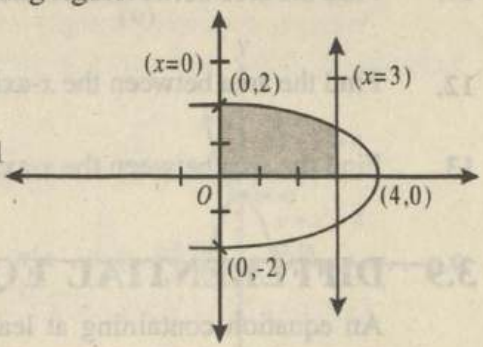
$$= \frac{16}{3} - \left(\frac{-16}{3}\right) = \frac{32}{3}$$

Example 5: Find the area between the x-axis and the curve $y^2 = 4 - x$ in the first quadrant from $x = 0$ to $x = 3$.

Solution: The branch of the curve above the x-axis is

$$y = \sqrt{4-x}$$

The area to be determined is shaded in the adjoining figure.



Thus the required area $= \int_0^3 \sqrt{4-x}\, dx$

Let $4 - x = t$ (i), then $-dx = dt \Rightarrow dx = -dt$

Putting $x = 0$ and $x = 3$ (i), we get $t = 4$ and $t = 1$

Now the required area $= \int_4^1 t^{\frac{1}{2}} \times (-dt) = \int_4^1 t^{\frac{1}{2}}\, dt$

$$= \int_1^4 t^{\frac{1}{2}}\, dt = \left| \frac{t^{\frac{3}{2}}}{\frac{3}{2}} \right|_1^4$$

$$= \frac{2}{3} \left| t^{\frac{3}{2}} \right|_1^4 = \frac{2}{3} \left[(4)^{\frac{3}{2}} - (1)^{\frac{3}{2}} \right] = \frac{2}{3}[8-1] = \frac{14}{3} \text{ (square units)}$$

EXERCISE 3.7

1. Find the area between the x-axis and the curve $y = x^2 + 1$ from $x = 1$ to $x = 2$.
2. Find the area, above the x-axis and under the curve $y = 5 - x^2$ from $x = -1$ to $x = 2$.
3. Find the area below the curve $y = 3\sqrt{x}$ and above the x-axis between $x = 1$ and $x = 4$.
4. Find the area bounded by cos function from $x = -\frac{\pi}{2}$ to $x = \frac{\pi}{2}$.
5. Find the area between the x-axis and the curve $y = 4x - x^2$.
6. Determine the area bounded by the parabola $y = x^2 + 2x - 3$ and the x-axis.
7. Find the area bounded by the curve $y = x^3 + 1$, the x-axis and line $x = 2$.
8. Find the area bounded by the curve $y = x^3 - 4x$ and the x-axis.

9. Find the area between the curve $y = x(x-1)(x+1)$ and the x-axis.

10. Find the area above the x-axis, bounded by the curve $y^2 = 3 - x$ from $x = -1$ to $x = 2$.

11. Find the area between the x-axis and the curve $y = cos\frac{1}{2}x$ from $x = -\pi$ to π.

12. Find the area between the x-axis and the curve $y = sin 2x$ from $x = 0$ to $x = \frac{\pi}{3}$.

13. Find the area between the x-axis and the curve $y = \sqrt{2ax - x^2}$ when $a > 0$.

3.9 DIFFERENTIAL EQUATIONS

An equation containing at least one derivative of a dependent variable with respect to an independent variable such as

$$y\frac{dy}{dx} + 2x = 0 \qquad \text{(i)}$$

or $$\frac{x\,d^2y}{dx^2} + \frac{dy}{dx} - 2x = 0 \qquad \text{(ii)}$$

is called a differential equation.

Derivatives may be of first or higher orders. A differential equation containing only derivative of first order can be written in terms of differentials. So we can write the equation (i) as $y\,dy + 2x\,dx = 0$ but the equation (ii) cannot be written in terms of differentials.

Order: The **order of a differential equation** is the **order of the highest derivative in the equation**. As the order of the equation (i) is one so it is called a **first order differential** equation. But equation (ii) contains the second order derivative and is called a second order differential equation.

3.9.1 Solution of a Differential Equation of First Order

Consider the equation

$$y = Ax^2 + 4 \qquad \text{(iii)}$$

where A is a real constant

Differentiating (iii) with respect to x gives

$$\frac{dy}{dx} = 2Ax \qquad \text{(iv)}$$

From (iii) $A = \frac{y-4}{x^2}$, so putting the value of A in (iv), we get

EXERCISE 3.8

1. Check that each of the following equations written against the differential equation is its solution.

(i) $x\frac{dy}{dx} = 1 + y$, $y = cx - 1$

(ii) $x^2(2y+1)\frac{dy}{dx} - 1 = 0$, $y^2 + y = c - \frac{1}{x}$

(iii) $y\frac{dy}{dx} - e^{2x} = 1$, $y^2 = e^{2x} + 2x + c$

(iv) $\frac{1}{x}\frac{dy}{dx} - 2y = 0$, $y = ce^{x^2}$

(v) $\frac{dy}{dx} = \frac{y^2+1}{e^{-x}}$, $y = \tan(e^x + c)$

Solve the following differential equations:

2. $\frac{dy}{dx} = -y$

3. $y\,dx + x\,dy = 0$

4. $\frac{dy}{dx} = \frac{1-x}{y}$

5. $\frac{dy}{dx} = \frac{y}{x^2}$, $(y > 0)$

6. $\sin y \operatorname{cosec} x \frac{dy}{dx} = 1$

7. $x\,dy + y(x-1)\,dx = 0$

8. $\frac{x^2+1}{y+1} = \frac{x}{y}\cdot\frac{dy}{dx}$, $(x, y > 0)$

9. $\frac{1}{x}\frac{dy}{dx} = \frac{1}{2}(1+y^2)$

10. $2x^2 y\frac{dy}{dx} = x^2 - 1$

11. $\frac{dy}{dx} + \frac{2xy}{2y+1} = x$

12. $(x^2 - yx^2)\frac{dy}{dx} + y^2 + xy^2 = 0$

13. $\sec^2 x \tan y\,dx + \sec^2 y \tan x\,dy = 0$

14. $\left(y - x\frac{dy}{dx}\right) = 2\left(y^2 + \frac{dy}{dx}\right)$

15. $1 + \cos x \tan y \dfrac{dy}{dx} = 0$

16. $y - x\dfrac{dy}{dx} = 3\left(1 + x\dfrac{dy}{dx}\right)$

17. $\sec x + \tan y \dfrac{dy}{dx} = 0$

18. $(e^x + e^{-x})\dfrac{dy}{dx} = e^x - e^{-x}$

19. Find the general solution of the equation $\dfrac{dy}{dx} - x = xy^2$

Also find the particular solution if $y = 1$ when $x = 0$.

20. Solve the differential equation $\dfrac{dx}{dt} = 2x$ given that $x = 4$ when $t = 0$.

21. Solve the differential equation $\dfrac{ds}{dt} + 2st = 0$. Also find the particular solution if $s = 4e$, when $t = 0$.

22. In a culture, bacteria increases at the rate proportional to the number of bacteria present. If bacteria are 200 initially and are doubled in 2 hours, find the number of bacteria present four hours later.

23. A ball is thrown vertically upward with a velocity of 2450 cm/sec. Neglecting air resistance, find

(i) velocity of ball at any time t

(ii) distance traveled in any time t

(iii) maximum height attained by the ball.

Thus I divides AD in the ratio $c:\frac{ac}{b+c}$ or in the ratio $b+c:a$

Coordinates of I are

$$\left(\frac{(b+c)\frac{bx_2+cx_3}{b+c}+ax_1}{a+b+c}, \frac{(b+c)\frac{by_2+cy_3}{b+c}+ay_1}{a+b+c}\right)$$

i.e., $$\left(\frac{ax_1+bx_2+cx_3}{a+b+c}, \frac{ay_1+by_2+cy_3}{a+b+c}\right)$$

The symmetry of these coordinates shows that the bisector of $\angle C$ will also pass through this point.

Thus the angle bisectors of a triangle are concurrent.

EXERCISE 4.1

1. Describe the location in the plane of the point $P(x, y)$ for which

(i) $x>0$ (ii) $x>0$ and $y>0$ (iii) $x=0$

(iv) $y=0$ (v) $x<0$ and $y\geq 0$ (vi) $x=y$

(vii) $|x|=-|y|$ (viii) $|x|\geq 3$ (ix) $x>2$ and $y=2$

(x) x and y have opposite signs.

2. Find in each of the following

(i) the distance between the two given points

(ii) midpoint of the line segment joining the two points

(a) $A(3, 1); B(-2,-4)$ (b) $A(-8, 3); B(2, -1)$ (c) $A\left(-\sqrt{5},-\frac{1}{3}\right); B\left(-3\sqrt{5},5\right)$

3. Which of the following points are at a distance of 15 units from the origin?

(a) $\left(\sqrt{176},7\right)$ (b) $(10, -10)$ (c) $(1, 15)$ (d) $\left(\frac{15}{2},\frac{15}{2}\right)$

4. Show that

(i) the points $A(0, 2), B\left(\sqrt{3},-1\right)$ and $C(0,-2)$ are vertices of a right triangle.

(ii) the points $A(3, 1)$, $B(-2,-3)$ and $C(2, 2)$ are vertices of an isosceles triangle.

(iii) the points $A(5, 2)$, $B(-2, 3)$, $C(-3,-4)$ and $D(4, -5)$ are vertices of a parallelogram. Is the parallelogram a square?

5. The midpoints of the sides of a triangle are $(1, -1)$, $(-4, -3)$ and $(-1, 1)$. Find coordinates of the vertices of the triangle.

6. Find h such that the points $A(\sqrt{3},-1)$, $B(0, 2)$ and $C(h, -2)$ are vertices of a right triangle with right angle at the vertex A.

7. Find h such that $A(-1, h)$, $B(3, 2)$ and $C(7, 3)$ are collinear.

8. The points $A(-5,-2)$ and $B(5,-4)$ are ends of a diameter of a circle. Find the centre and radius of the circle.

9. Find h such that the points $A(h,1)$, $B(2,7)$ and $C(-6,-7)$ are vertices of a right triangle with right angle at the vertex A.

10. A quadrilateral has the points $A(9, 3)$, $B(-7, 7)$, $C(-3,-7)$ and $D(5, -5)$ as its vertices. Find the midpoints of its sides. Show that the figure formed by joining the midpoints consecutively is a parallelogram.

11. Find h such that the quadrilateral with vertices $A(-3, 0)$, $B(1, -2)$, $C(5, 0)$ and $D(1, h)$ is parallelogram. Is it a square?

12. If two vertices of an equilateral triangle are $A(-3, 0)$ and $B(3, 0)$, find the third vertex. How many of these triangles are possible?

13. Find the points trisecting the join of $A(-1, 4)$ and $B(6, 2)$.

14. Find the point three–fifth of the way along the line segment from $A(-5, 8)$ to $B(5, 3)$.

15. Find the point P on the joint of $A(1, 4)$ and $B(5, 6)$ that is twice as far from A as B is from A and lies

(i) on the same side of A as B does. (ii) on the opposite side of A as B does.

16. Find the point which is equidistant from the points $A(5, 3)$, $B(-2, 2)$ and $C(4, 2)$. What is the radius of the circumcircle of the ΔABC?

17. The points $(4, -2)$, $(-2, 4)$ and $(5, 5)$ are the vertices of a triangle. Find in-centre of the triangle.

18. Find the points that divide the line segment joining $A(x_1, y_1)$ and $B(x_2, y_2)$ into four equal parts.

Example 3: The xy-coordinate axes are rotated about the origin through an angle of $30°$. If the xy-coordinates of a point are $(5, 7)$, find its XY-coordinates, where OX and OY are the axes obtained after rotation.

Solution: Let (X, Y) be the coordinates of P referred to the XY-axes. Here $\theta = 30°$. From equations (3) above, we have

$$X = 5\cos 30° + 7\sin 30° \text{ and } Y = -5\sin 30° + 7\cos 30°$$

$$\Rightarrow \quad X = \frac{5\sqrt{3}}{2} + \frac{7}{2} \text{ and } Y = \frac{-5}{2} + \frac{7\sqrt{3}}{2}$$

$$\text{i.e., } (X, Y) = \left(\frac{5\sqrt{3}+7}{2}, \frac{-5+7\sqrt{3}}{2}\right)$$

are the required coordinates.

Example 4: The xy-axes are rotated about the origin through an angle of $\arctan\frac{4}{3}$ lying in the first quadrant. The coordinates of a point P referred to the new axes OX and OY are $P(-1, -7)$. Find the coordinates of P referred to the xy-coordinate system.

Solution: Let $P(x, y)$ be the coordinates of P referred to the xy-coordinate system. Angle of rotation is given by $\arctan\theta = \frac{4}{3}$. Therefore, $\sin\theta = \frac{4}{5}$, $\cos\theta = \frac{3}{5}$.

From equations (3) above, we have

$$X = x\cos\theta + y\sin\theta \text{ and } Y = -x\sin\theta + y\cos\theta$$

or $\quad -1 = \frac{3}{5}x + \frac{4}{5}y \quad$ and $\quad -7 = -\frac{4}{5}x + \frac{3}{5}y$

or $\quad 3x + 4y + 5 = 0 \Rightarrow -4x + 3y + 35 = 0$

Solving these equations, we have

$$\frac{x}{125} = \frac{y}{-125} = \frac{1}{25} \Rightarrow x = 5, \quad y = -5$$

Thus coordinates of P referred to the xy-system are $(5, -5)$.

EXERCISE 4.2

1. The two points P and O' are given in xy-coordinate system. Find the XY-coordinates of P referred to the translated axes $O'X$ and $O'Y$.

(i) $P(3, 2)$; $O'(1, 3)$ (ii) $P(-2, 6)$; $O'(-3, 2)$

(iii) $P(-6, -8)$; $O'(-4, -6)$ (iv) $P\left(\frac{3}{2}, \frac{5}{2}\right)$; $O'\left(-\frac{1}{2}, \frac{7}{2}\right)$

2. The xy-coordinate axes are translated through the point O' whose coordinates are given in xy-coordinate. The coordinates of P are given in the XY-coordinate system. Find the coordinates of P in xy-coordinate system.

(i) $P(8, 10)$; $O'(3, 4)$ (ii) $P(-5, -3)$; $O'(-2, -6)$

(iii) $P\left(-\frac{3}{4}, -\frac{7}{6}\right)$; $O'\left(\frac{1}{4}, -\frac{1}{6}\right)$ (iv) $P(4, -3)$; $O'(-2, 3)$

3. The xy-coordinate axes are rotated about the origin through the indicated angle. The new axes are OX and OY. Find the XY-coordinates of the point P with the given xy-coordinates.

(i) $P(5, 3)$; $\theta = 45°$ (ii) $P(3, -7)$; $\theta = 30°$

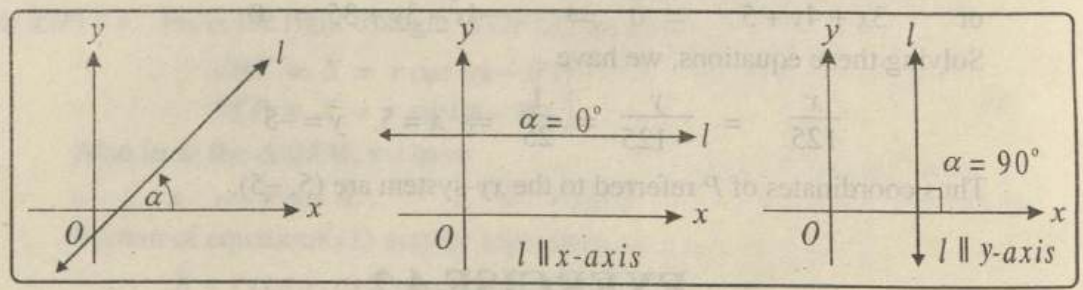
(iii) $P(11, -15)$; $\theta = 60°$ (iv) $P(15, 10)$; $\theta = \arctan\frac{1}{3}$

4. The xy-coordinate axes are rotated about the origin through the indicated angle and the new axes are OX and OY. Find the xy-coordinates of P with the given XY-coordinates.

(i) $P(-5, 3)$; $\theta = 30°$ (ii) $P(-7\sqrt{2}, 5\sqrt{2})$; $\theta = 45°$

4.3 EQUATIONS OF STRAIGHT LINES

Inclination of a Line: The angle $\alpha (0° < \alpha < 180°)$ measured counterclockwise from positive x-axis to a non-horizontal straight line l is called the **inclination** of l.



Observe that the angle α in the different positions of the line l are α, $0°$ and $90°$ respectively.

Note: (i) If l is parallel to x-axis, then $\alpha = 0°$

(ii) If l is parallel to y-axis, then $\alpha = 90°$

$= \frac{1}{2}[-2c(a+a)]$, expanding by the second row

$= -2ca$

Thus $\Delta = 2ac$

Example 6: By considering the area of the region bounded by the triangle with vertices

$A(1,4)$, $B(2,-3)$ and $C(3,-10)$,

check whether the three points are collinear or not.

Solution: Area Δ of the region bounded by the triangle ABC is,

$$\Delta = \frac{1}{2}\begin{vmatrix} 1 & 4 & 1 \\ 2 & -3 & 1 \\ 3 & -10 & 1 \end{vmatrix} = \frac{1}{2}\begin{vmatrix} 1 & 4 & 1 \\ 1 & -7 & 0 \\ 2 & -14 & 0 \end{vmatrix} \text{ by } R_2 - R_1 \text{ and } R_3 - R_1$$

$= \frac{1}{2}[1(-14+14)]$, expanding by third column

$= 0$

Thus the points are collinear.

EXERCISE 4.3

1. Find the slope and inclination of the line joining the points:
   (i) $(-2,4)$;$(5,11)$ (ii) $(3,-2)$; $(2,7)$ (iii)$(4,6)$; $(4,8)$
   Sketch each line in the plane.
2. In the triangle $A(8,6)$,$B(-4,2)$, $C(-2,-6)$, find the slope of
   (i) each side of the triangle (ii) each median of the triangle
   (iii) each altitude of the triangle
3. By means of slopes, show that the following points lie on the same line:
   (a) $(-1,-3)$; $(1,5)$; $(2,9)$ (b) $(4,-5)$; $(7,5)$; $(10,15)$
   (c) $(-4,6)$; $(3,8)$; $(10,10)$ (d) $(a, 2b)$; $(c, a+b)$; $(2c-a, 2a)$
4. Find k so that the line joining $A(7,3)$; $B(k,-6)$ and the line joining $C(-4,5)$; $D(-6,4)$ are (i) parallel (ii) perpendicular.
5. Using slopes, show that the triangle with its vertices $A(6,1)$, $B(2,7)$ and $C(-6,-7)$ is a right triangle.
6. The three points $A(7,-1)$, $B(-2,2)$ and $C(1,4)$ are consecutive vertices of a parallelogram. Find the fourth vertex.

7. The points $A(-1, 2)$, $B(3,-1)$ and $C(6,3)$ are consecutive vertices of a rhombus. Find the fourth vertex and show that the diagonals of the rhombus are perpendicular to each other.

8. Two pairs of points are given. Find whether the two lines determined by these points are :
   (i) parallel (ii) perpendicular (iii) none
   (a) $(1,-2)$, $(2, 4)$ and $(4, 1)$, $(-8,2)$ **(b)** $(-3, 4)$, $(6, 2)$ and $(4, 5)$, $(-2,-7)$

9. Find an equation of
   (a) the horizontal line through $(7, -9)$
   (b) the vertical line through $(-5, 3)$
   (c) the line bisecting the first and third quadrants.
   (d) the line bisecting the second and fourth quadrants.

10. Find an equation of the line
   (a) through $A(-6, 5)$ having slope 7 (b) through $(8, -3)$ having slope 0
   (c) through $(-8, 5)$ having slope undefined
   (d) through $(-5,-3)$ and $(9, -1)$ (e) y-intercept -7 and slope -5
   (f) x-intercept:-3 and y-intercept:4 (g) x-intercept:-9 and slope :-4

11. Find an equation of the perpendicular bisector of the segment joining the points $A(3, 5)$ and $B(9,8)$

12. Find equations of the sides, altitudes and medians of the triangle whose vertices are $A(-3, 2)$, $B(5, 4)$ and $C(3, -8)$.

13. Find an equation of the line through $(-4,-6)$ and perpendicular to a line having slope $\frac{-3}{2}$.

14. Find an equation of the line through $(11,-5)$ and parallel to a line with slope -24.

15. The points $A(-1, 2)$, $B(6, 3)$ and $C(2,-4)$ are vertices of a triangle. Show that the line joining the midpoint D of AB and the midpoint E of AC is parallel to BC and $DE = \frac{1}{2} BC$.

16. A milkman can sell 560 litres of milk at Rs.12.50 per litre and 700 litres of milk at Rs. 12.00 per litre. Assuming the graph of the sale price and the milk sold to be a straight line, find the number of litres of milk that the milkman can sell at Rs.12.25 per litre.

17. The population of Pakistan to the nearest million was 60 million in 1961 and 95 million in 1981. Using t as the number of years after 1961, find an equation of the line that gives the population in terms of t. Use this equation to find the population in (a) 1947 (b) 1997.

18. A house was purchased for Rs.1 million in 1980. It is worth Rs.4 million in 1996. Assuming that the value increased by the same amount each year, find an equation that gives the value of the house after t years of the date of purchase. What was its value in 1990?

19. Plot the Celsius (C) and Fahrenheit (F) temperature scales on the horizontal axis and the vertical axis respectively. Draw the line joining the freezing point and the boiling point of water. Find an equation giving F temperature in terms of C.

20. The average entry test score of engineering candidates was 592 in the year 1998 while the score was 564 in 2002. Assuming that the relationship between time and score is linear, find the average score for 2006.

21. Convert each of the following equation into

(i) Slope intercept form (ii) two intercepts form (iii) normal form

(a) $2x-4y+11=0$ (b) $4x+7y-2=0$ (c) $15y-8x+3=0$

Also find the length of the perpendicular from (0, 0) to each line.

22. In each of the following check whether the two lines are

(i) parallel (ii) perpendicular (iii) neither parallel nor perpendicular

(a) $2x+y-3=0$; $4x+2y+5=0$

(b) $3y=2x+5$; $3x+2y-8=0$

(c) $4y+2x-1=0$; $x-2y-7=0$

(d) $4x-y+2=0$; $12x-3y+1=0$

(e) $12x+35y-7=0$; $105x-36y+11=0$

23. Find the distance between the given parallel lines. Sketch the lines. Also find an equation of the parallel line lying midway between them.

(a) $3x-4y+3=0$; $3x-4y+7=0$

(b) $12x+5y-6=0$; $12x+5y+13=0$

(c) $x+2y-5=0$; $2x+4y=1$

24. Find an equation of the line through $(-4, 7)$ and parallel to the line $2x-7y+4=0$.

25. Find an equation of the line through $(5,-8)$ and perpendicular to the join of $A(-15,-8)$, $B(10,7)$.

26. Find equations of two parallel lines perpendicular to $2x-y+3=0$ such that the product of the x- and y-intercepts of each is 3.

27. One vertex of a parallelogram is (1, 4), the diagonal intersect at (2, 1) and the sides have slopes 1 and $\frac{-1}{7}$. Find the other three vertices.

28. Find whether the given point lies above or below the given line

(a) $(5, 8)$; $2x-3y+6=0$

(b) $(-7, 6)$; $4x+3y-9=0$

29. Check whether the given points are on the same or opposite sides of the given line.

(a) $(0, 0)$ and $(-4, 7)$; $6x-7y+70=0$

(b) $(2, 3)$ and $(-2, 3)$; $3x-5y+8=0$

30. Find the distance from the point $P(6,-1)$ to the line $6x-4y+9=0$.

31. Find the area of the triangular region whose vertices are $A(5, 3)$, $B(-2, 2)$, $C(4, 2)$.

32. The coordinates of three points are $A(2, 3)$, $B(-1, 1)$ and $C(4, -5)$. By computing the area bounded by ABC check whether the points are collinear.

4.5. ANGLE BETWEEN TWO LINES

Let l_1 and l_2 be two intersecting lines, which meet at a point P. At the point P two supplementary angles are formed, by the lines l_1 and l_2.

Unless $l_1 \perp l_2$ one of the two angles is acute. The angle from l_1 to l_2 is the angle θ through which l_1 is rotated anti-clockwise about the point P so that it coincides with l_2.

In the figure below θ is angle of intersection of the two lines and it is measured from l_1 to l_2 in counterclockwise direction, ψ is also angle of intersection but it is measured from l_2 to l_1.

With this convention for angle of intersection, it is clear that the inclination of a line is the angle measured in the counterclockwise direction from the positive x-axis to the line and it tallies with the earlier definition of the inclination of a line.

EXERCISE 4.4

1. Find the point of intersection of the lines

(i) $x-2y+1=0$ and $2x-y+2=0$

(ii) $3x+y+12=0$ and $x+2y-1=0$

(iii) $x+4y-12=0$ and $x-3y+3=0$

2. Find an equation of the line through

(i) the point $(2,-9)$ and the intersection of the lines

$2x+5y-8=0$ and $3x-4y-6=0$

(ii) the intersection of the lines

$x-y-4=0$ and $7x+y+20$ and

(a) parallel (b) perpendicular

to the line $6x+y-14=0$

(iii) through the intersection of the lines $x+2y+3=0$, $3x+4y+7=0$ and making equal intercepts on the axes.

3. Find an equation of the line through the intersection of

$16x-10y-33=0$; $12x+14y+29=0$ and the intersection of

$x-y+4=0$; $x-7y+2=0$

4. Find the condition that the lines $y=m_1x+c_1; y=m_2x+c_2$ and $y=m_3x+c_3$ are concurrent.

5. Determine the value of p such that the lines $2x-3y-1=0$, $3x-y-5=0$ and $3x+py+8=0$ meet at a point.

6. Show that the lines $4x-3y-8=0$, $3x-4y-6=0$ and $x-y-2=0$ are concurrent and the third-line bisects the angle formed by the first two lines.

7. The vertices of a triangle are $A(-2,3)$, $B(-4,1)$ and $C(3,5)$. Find coordinates of the

(i) centroid (ii) orthocentre (iii) circumcentre of the triangle

Are these three points collinear?

8. Check whether the lines

$4x-3y-8=0$; $3x-4y-6=0$; $x-y-2=0$

are concurrent. If so, find the point where they meet.

9. Find the coordinates of the vertices of the triangle formed by the lines

$x-2y-6=0$; $3x-y+3=0$; $2x+y-4=0$

Also find measures of the angles of the triangle.

10. Find the angle measured from the line l_1 to the line l_2 where

(a) l_1 : Joining (2,7) and (7, 10)
l_2 : Joining (1, 1) and (−5, 3)

(b) l_1 : Joining (3, −1) and (5, 7)
l_2 : Joining (2, 4) and (−8, 2)

(c) l_1 : Joining (1, −7) and (6, −4)
l_2 : Joining (−1, 2) and (−6, −1)

(d) l_1 : Joining (−9, −1) and (3, −5)
l_2 : Joining (2, 7) and (−6, −7)

Also find the acute angle in each case.

11. Find the interior angles of the triangle whose vertices are

(a) $A(-2,11), B(-6,-3), (4,-9)$ (b) $A(6, 1), B(2, 7), C(-6,-7)$

(c) $A(2,-5), B(-4,-3), (-1, 5)$ (d) $A(2, 8), B(-5, 4), C(4,-9)$

12. Find the interior angles of the quadrilateral whose vertices are $A(5, 2)$, $B(-2,3)$, $C(-3,-4)$ and $D(4,-5)$

13. Show that the points
$A(0, 0), B(2, 1), C(3, 3), D(1, 2)$ are the vertices of a rhombus and find its interior angles.

14. Find the area of the region bounded by the triangle whose sides are

$7x-y-10=0$; $10x+y-41=0$; $3x+2y+3=0$

15. The vertices of a triangle are $A(-2, 3)$, $B(-4, 1)$ and $C(3, 5)$. Find the centre of the circum centre of the triangle.

16. Express the given system of equations in matrix form. Find in each case whether the lines are concurrent.

(a) $x+3y-2=0$; $2x-y+4=0$; $x-11y+14=0$

(b) $2x+3y+4=0$; $x-2y-3=0$; $3x+y-8=0$

(c) $3x-4y-2=0$, $x+2y-4=0$, $3x-2y+5=0$.

17. Find a system of linear equations corresponding to the given matrix form. Check whether the lines represented by the system are concurrent.

(a) $\begin{bmatrix} 1 & 0 & -1 \\ 2 & 0 & 1 \\ 0 & -1 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

(b) $\begin{bmatrix} 1 & 1 & 2 \\ 2 & 4 & -3 \\ 3 & 6 & -5 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

4.6 HOMOGENEOUS EQUATION OF THE SECOND DEGREE IN TWO VARIABLES

We have already seen that if a graph is a straight line, then its equation is a linear equation in the variables x and y. Conversely, the graph of any linear equation in x and y is a straight line.

Example 3: Find a joint equation of the straight lines through the origin perpendicular to the lines represented by

$$x^2 + xy - 6y^2 = 0 \quad (1)$$

Solution: (1) may be written as

$$(x-2y)(x+3y) = 0$$

Thus the lines represented by (1) are

$$x - 2y = 0 \quad (2)$$

and $$x + 3y = 0 \quad (3)$$

The line through (0, 0) and perpendicular to (2) is

$$y = -2x \quad \text{or} \quad y + 2x = 0 \quad (4)$$

Similarly, the line through (0, 0) and perpendicular to (3) is

$$y = 3x \quad \text{or} \quad y - 3x = 0 \quad (5)$$

Joint equation of the lines (4) and (5) is

$$(y+2x)(y-3x) = 0 \quad \text{or} \quad y^2 - xy - 6x^2 = 0$$

EXERCISE 4.5

Find the lines represented by each of the following and also find measure of the angle between them **(Problems 1-6):**

1. $10x^2 - 23xy - 5y^2 = 0$

2. $3x^2 + 7xy + 2y^2 = 0$

3. $9x^2 + 24xy + 16y^2 = 0$

4. $2x^2 + 3xy - 5y^2 = 0$

5. $6x^2 - 19xy + 15y^2 = 0$

6. $x^2 - 2xy \sec\alpha + y^2 = 0$

7. Find a joint equation of the lines through the origin and perpendicular to the lines:

$$x^2 - 2xy \tan\alpha - y^2 = 0$$

8. Find a joint equation of the lines through the origin and perpendicular to the lines:

$$ax^2 + 2hxy + by^2 = 0$$

9. Find the area of the region bounded by:

$$10x^2 - xy - 21y^2 = 0 \text{ and } x + y + 1 = 0$$

(ii) See figure 5.33(a) for the graphs of the inequalities $2x + y \geq 2$ and $x + 2y \leq 10$.

The graph of $x \geq 0$ consists of the open half-plane to the right of the corresponding line $x = 0$ (y-axis) of the inequality $x \geq 0$ and its graph. See figure 5.34(a).

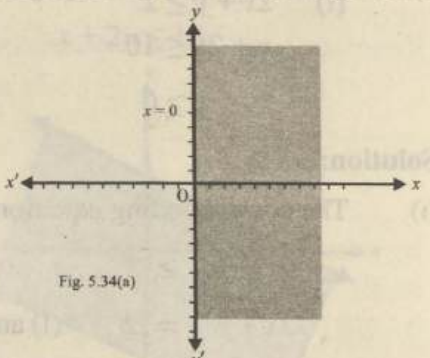


Fig. 5.34(a)

Thus the solution region of the inequalities in (ii) is partially shown in figure 5.34(b). This region is the intersection of graphs in figure 5.33(a) and 5.34(a).

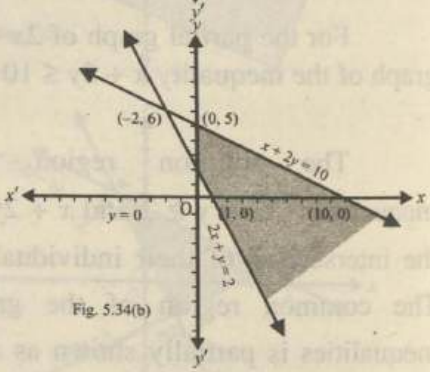


Fig. 5.34(b)

(iii) The graphs of the system of inequalities in (iii) are drawn in the solution of (i) and (ii). The solution region in this case, is shown as shaded region *ABCD* in figure 5.34. (c).

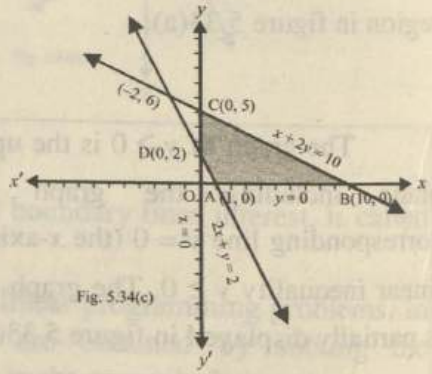


Fig. 5.34(c)

EXERCISE 5.1

1. Graph the solution set of each of the following linear inequality in xy-plane:

(i) $2x + y \leq 6$ (ii) $3x + 7y \geq 21$ (iii) $3x - 2y \geq 6$

(iv) $5x - 4y \leq 20$ (v) $2x + 1 \geq 0$ (vi) $3y - 4 \leq 0$

2. Indicate the solution set of the following systems of linear inequalities by shading:

(i) $2x - 3y \leq 6$, $2x + 3y \leq 12$ (ii) $x + y \geq 5$, $-y + x \leq 1$ (iii) $3x + 7y \geq 21$, $x - y \leq 2$

(iv) $4x - 3y \leq 12$
$x \geq -\frac{3}{2}$

(v) $3x + 7y \geq 21$
$y \leq 4$

3. Indicate the solution region of the following systems of linear inequalities by shading:

(i) $2x - 3y \leq 6$
$2x + 3y \leq 12$
$y \geq 0$

(ii) $x + y \leq 5$
$y - 2x \leq 2$
$x \geq 0$

(iii) $x + y \geq 5$
$x - y \geq 1$
$y \geq 0$

(iv) $3x + 7y \leq 21$
$x - y \leq 2$
$x \geq 0$

(v) $3x + 7y \leq 21$
$x - y \leq 2$
$y \geq 0$

(vi) $3x + 7y \leq 21$
$2x - y \geq -3$
$x \geq 0$

4. Graph the solution region of the following system of linear inequalities and find the corner points in each case.

(i) $2x - 3y \leq 6$
$2x + 3y \leq 12$
$x \geq 0$

(ii) $x + y \leq 5$
$-2x + y \leq 2$
$y \geq 0$

(iii) $3x + 7y \leq 21$
$2x - y \leq -3$
$y \geq 0$

(iv) $3x + 2y \geq 6$
$x + 3y \leq 6$
$y \geq 0$

(v) $5x + 7y \leq 35$
$-x + 3y \leq 3$
$x \geq 0$

(vi) $5x + 7y \leq 35$
$x - 2y \leq 4$
$x \geq 0$

5. Graph the solution region of the following system of linear inequalities by shading.

(i) $3x - 4y \leq 12$
$3x + 2y \geq 3$
$x + 2y \leq 9$

(ii) $3x - 4y \leq 12$
$x + 2y \leq 6$
$x + y \geq 1$

(iii) $2x + y \leq 4$
$2x - 3y \geq 12$
$x + 2y \leq 6$

(iv) $2x + y \leq 10$
$x + y \leq 7$
$-2x + y \leq 4$

(v) $2x + 3y \leq 18$
$2x + y \leq 10$
$-2x + y \leq 2$

(vi) $3x - 2y \geq 3$
$x + 4y \leq 12$
$3x + y \leq 12$

5.4 PROBLEM CONSTRAINTS

In the beginning we described that linear inequalities prescribe limitations and restrictions on allocation of available sources. While tackling a certain problem from every day life each linear inequality concerning the problem is named as **problem constraint**. The system of linear inequalities involved in the problem concerned are called **problem constraints**. The variables used in the system of linear inequalities relating to the problems of every day life are non-negative and are called **non-**

(b) The graph of system $x+2y \le 8$, $x \ge 0$, $y \ge 0$ is a triangular region indicated in figure 5.45(d).

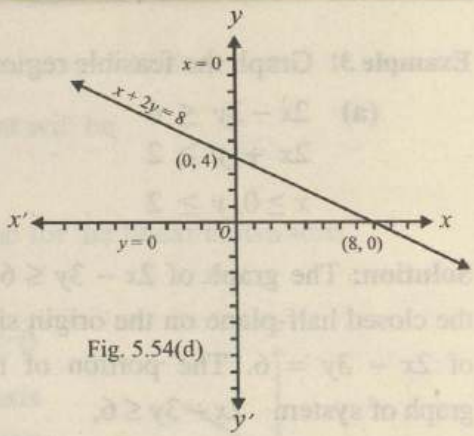


Fig. 5.54(d)

Thus the graph of the system

$$2x - 3y \le 6$$
$$2x + y \ge 2$$
$$x + 2y \le 8, \quad x \ge 0, \ y \ge 0$$

is the intersection of the graphs shown in figures 5.54(c) and 5.54(d). It is the indicated in the figure 5.54(e).

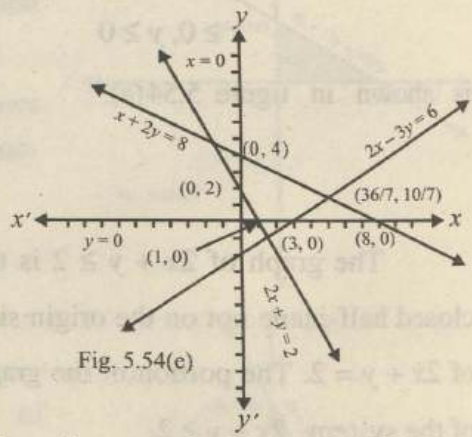
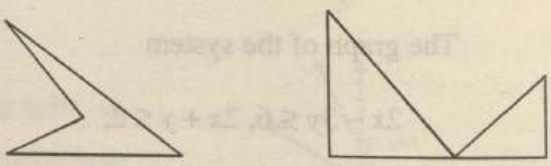


Fig. 5.54(e)

Note: The corner points of feasible region for the set of constraints in (a) are (1, 0) (3, 0) and (0, 2) while the corner points of the feasible region for the set of constraints in (b) are (1, 0), (3, 0), $\left(\frac{36}{7}, \frac{10}{7}\right)$, (0, 4) and (0, 2).

We see that the feasible solution regions in example 3(a) and 3(b) are of different types. The feasible region in example 3(a) is unbounded as it cannot be enclosed in any circle how large it may be while the feasible region in example 3(b) can easily be enclosed within a circle, so it is bounded. If the line segment obtained by joining any two points of a region lies entirely within the region, then the region is called **convex**.

Both the feasible regions of example 3(a) and 3(b) are convex but the regions such as shown in the adjoining figures are not convex.



EXERCISE 5.2

1. Graph the feasible region of the following system of linear inequalities and find the corner points in each case.

(i) $2x - 3y \le 6$, $2x + 3y \le 12$, $x \ge 0, y \ge 0$

(ii) $x + y \le 5$, $-2x + y \le 2$, $x \ge 0, y \ge 0$

(iii) $x + y \le 5$, $-2x + y \ge 2$, $x \ge 0$

(iv) $3x + 7y \leq 21$, $x - y \leq 3$, $x \geq 0, y \geq 0$

(v) $3x + 2y \geq 6$, $x + y \leq 4$, $x \geq 0, y \geq 0$

(vi) $5x + 7y \leq 35$, $x - 2y \leq 4$, $x \geq 0, y \geq 0$

2. Graph the feasible region of the following system of linear inequalities and find the corner points in each case.

(i) $2x + y \leq 10$, $x + 4y \leq 12$, $x + 2y \leq 10$, $x \geq 0, y \geq 0$

(ii) $2x + 3y \leq 18$, $2x + y \leq 10$, $x + 4y \leq 12$, $x \geq 0, y \geq 0$

(iii) $2x + 3y \leq 18$, $x + 4y \leq 12$, $3x + y \leq 12$, $x \geq 0, y \geq 0$

(iv) $x + 2y \leq 14$, $3x + 4y \leq 36$, $2x + y \leq 10$, $x \geq 0, y \geq 0$

(v) $x + 3y \leq 15$, $2x + y \leq 12$, $4x + 3y \leq 24$, $x \geq 0, y \geq 0$

(vi) $2x + y \leq 20$, $8x + 15y \leq 120$, $x + y \leq 11$, $x \geq 0, y \geq 0$

5.6 LINEAR PROGRAMMING

A function which is to be maximized or minimized is called an **objective function**. Note that there are infinitely many feasible solutions in the feasible region. The feasible solution which maximizes or minimizes the objective function is called the **optimal solution**. The theorem of linear programming states that the maximum and minimum values of the objective function occur at corner points of the feasible region.

Procedure for determining optimal solution:

(i) Graph the solution set of linear inequality constraints to determine feasible region.

(ii) Find the corner points of the feasible region.

(iii) Evaluate the objective function at each corner point to find the optimal solution.

Example 1: Find the maximum and minimum values of the function defined as:

$f(x, y) = 2x + 3y$ subject to the constraints;

$x - y \leq 2$, $x + y \leq 4$, $2x - y \leq 6$, $x \geq 0$

Solution: The graphs of $x - y \leq 2$ is the closed half-plane on the origin side of $x - y = 2$ and the graph of $x + y \leq 4$ is the closed half-plane not on the origin side of $x + y = 4$. The graph of the system

$$x - y \leq 2, x + y \geq 4$$

including the non-negative constraints $x \geq 0$ is partially displayed in the figure 5.61. The graph of $2x - y \leq 6$ consists of the graph of the line $2x - y = 6$ and the half plane on the origin side of the line $2x - y = 6$. A portion of the solution of the given system of inequalities is in the figure 5.62.

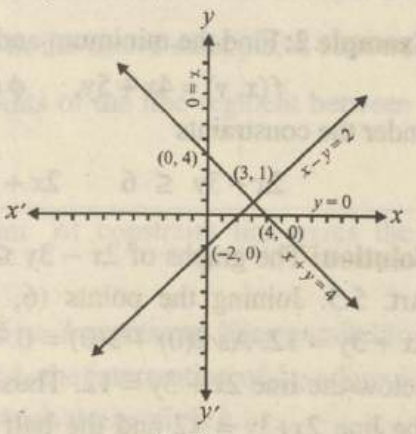


Fig. 5.61

EXERCISE 5.3

1. Maximize $f(x, y) = 2x + 5y$
subject to the constraints
$2y - x \leq 8$; $x - y \leq 4$; $x \geq 0$; $y \geq 0$

2. Maximize $f(x, y) = x + 3y$
subject to the constraints
$2x + 5y \leq 30$; $5x + 4y \leq 20$; $x \geq 0$; $y \geq 0$

3. Maximize $z = 2x + 3y$; subject to the constraints:
$3x + 4y \leq 12$; $2x + y \leq 4$; $2x - y \leq 4$; $x \geq 0$ $y \geq 0$

4. Minimize $z = 2x + y$; subject to the constraints:
$x + y \geq 3$; $7x + 5y \leq 35$; $x \geq 0$; $y \geq 0$

5. Maximize the function defined as; $f(x, y) = 2x + 3y$ subject to the constraints:
$2x + y \leq 8$; $x + 2y \leq 14$; $x \geq 0$; $y \geq 0$

6. Minimize $z = 3x + y$; subject to the constraints:
$3x + 5y \geq 15$; $x + 6y \geq 9$ $x \geq 0$; $y \geq 0$

7. Each unit of food X costs Rs.25 and contains 2 units of protein and 4 units of iron while each unit of food Y costs Rs.30 and contains 3 units of protein and 2 unit of iron. Each animal must receive at least 12 units of protein and 16 units of iron each day. How many units of each food should be fed to each animal at the minimum possible cost?

8. A dealer wishes to purchase a number of fans and sewing machines. He has only Rs. 5760 to invest and has space atmost for 20 items. A fan costs him Rs. 360 and a sewing machine costs Rs. 240. His expectation is that the can sell a fan at a profit of Rs. 22 and a sewing machine at a profit of Rs. 18. Assuming that he can sell all the items that he can buy, how should he invest his money in order to maximize his profit?

9. A machine can produce product A by using 2 units of chemical and 1 unit of a compound or can produce product B by using 1 unit of chemical and 2 units of the compound. Only 800 units of chemical and 1000 units of the compound are available. The profits per unit of A and B are Rs. 30 and Rs. 20 respectively, maximize the profit function.

Thus centres of the two circles are at $(0,0)$ and $(5,0)$

Radius of the first circle $=\sqrt{5}$; Radius of the second circle $=\sqrt{20}$

Equations of the circles are

$$x^2+y^2=5 \quad \text{and} \quad (x-5)^2+y^2=20$$

i.e., $$x^2+y^2=5 \quad \text{and} \quad x^2+y^2-10x+5=0$$

EXERCISE 6.1

1. In each of the following, find an equation of the circle with

(a) centre at $(5, -2)$ and radius 4

(b) centre at $(\sqrt{2},-3\sqrt{3})$ and radius $2\sqrt{2}$

(c) ends of a diameter at $(-3, 2)$ and $(5,-6)$.

2. Find the centre and radius of the circle with the given equation

(a) $x^2+y^2+12x-10y=0$

(b) $5x^2+5y^2+14x+12y-10=0$

(c) $x^2+y^2-6x+4y+13=0$

(d) $4x^2+4y^2-8x+12y-25=0$

3. Write an equation of the circle that passes through the given points

(a) $A(4, 5), B(-4,-3), C(8, -3)$

(b) $A(-7, 7), B(5,-1), C(10, 0)$

(c) $A(a, 0), B(0, b,), C(0, 0)$

(d) $A(5, 6), B(-3, 2), C(3, -4)$

4. In each of the following, find an equation of the circle passing through

(a) $A(3, -1), B(0, 1)$ and having centre at $4x-3y-3=0$

(b) $A(-3, 1)$ with radius 2 and centre at $2x-3y+3=0$

(c) $A(5, 1)$ and tangent to the line $2x-y-10=0$ at $B(3, -4)$

(d) $A(1, 4), B(-1, 8)$ and tangent to the line $x+3y-3=0$

5. Find an equation of a circle of radius a and lying in the second quadrant such that it is tangent to both the axes.

6. Show that the lines $3x-2y=0$ and $2x+3y-13=0$ are tangents to the circle $x^2+y^2+6x-4y=0$

7. Show that the circles

$x^2+y^2+2x-2y-7=0$ and $x^2+y^2-6x+4y-9=0$ touch externally.

8. Show that the circles

$x^2+y^2+2x-8=0$ and $x^2+y^2-6x+6y-46=0$ touch internally

9. Find equations of the circle of radius 2 and tangent to the line

$x-y-4=0$ at $A(1,-3)$.

6.2 TANGENTS AND NORMALS

A **tangent** to a curve is a line that touches the curve without cutting through it.

We know that for any curve whose equation is given by $y = f(x)$ or $f(x,y)=0$. the derivative $\frac{dy}{dx}$ is slope of the tangent at any point $P(x,y)$ to the curve. The equation of the tangent to the curve can easily be written by the point-slope formula. The **normal** to the curve at P is the line through P perpendicular to the tangent to the curve at P. This method can be very conveniently employed to find equations of tangent and normal to the circle $x^2+y^2+2gx+2fy+c=0$ at the point $P(x_1,y_1)$.

Here $f(x,y)=x^2+y^2+2gx+2fy+c=0$ (1)

Differentiating (1) w.r.t. x, we get

$$2x+2y\frac{dy}{dx}+2g+2f\frac{dy}{dx}=0 \quad \text{or} \quad \frac{dy}{dx}=-\frac{x+g}{y+f}$$

$$\left.\frac{dy}{dx}\right]_{(x_1,y_1)}=-\frac{x_1+g}{y_1+f}= \text{ Slope of the tangent at } (x_1,y_1)$$

Equation of the Tangent at P is given by

$$y-y_1=-\frac{x_1+g}{y_1+f}(x-x_1) \qquad \text{(Point-Slope form)}$$

or $y(y_1+f)-y_1^2-y_1f=-x(x_1+g)+x_1^2+x_1g$

or $xx_1+yy_1+gx+fy=x_1^2+y_1^2+gx_1+fy_1$

For $m = -2$: $-2x - y = 0$ or $2x + y = 0$

For $m = \frac{1}{2}$: $\frac{1}{2}x - y + \frac{5}{2} = 0$ or $x - 2y + 5 = 0$

Tangential distance $= \sqrt{1+4-4+4} = \sqrt{5}$.

Example 8: Tangents are drawn from $(-3, 4)$ to the circle $x^2 + y^2 = 21$. Find an equation of the line joining the points of contact (The line is called the **chord of contact**).

Solution: Let the points of contact of the two tangents be $P(x_1, y_1)$ and $Q(x_2, y_2)$

An equation of the tangent at P is

$$xx_1 + yy_1 = 21 \qquad (1)$$

An equation of the tangent at Q is

$$xx_2 + yy_2 = 21 \qquad (2)$$

Since (1) and (2) pass through $(-3,4)$, so

$$-3x_1 + 4y_1 = 21 \qquad (3)$$

and $$-3x_2 + 4y_2 = 21 \qquad (4)$$

(3) and (4) show that both the points $P(x_1, y_1)$, $Q(x_2, y_2)$ lie on $-3x + 4y = 21$ and so it is the required equation of the chord of contact.

EXERCISE 6.2

1. Write down equations of the tangent and normal to the circle

   (i) $x^2 + y^2 = 25$ at $(4, 3)$ and at $(5\cos\theta, 5\sin\theta)$

   (ii) $3x^2 + 3y^2 + 5x - 13y + 2 = 0$ at $\left(1, \frac{10}{3}\right)$

2. Write down equations of the tangent and normal to the circle

   $$4x^2 + 4y^2 - 16x + 24y - 117 = 0$$

   at the points on the circle whose abscissa is –4.

3 Check the position of the point (5 , 6) with respect to the circle

   (i) $x^2 + y^2 = 81$ (ii) $2x^2 + 2y^2 + 12x - 8y + 1 = 0$

4. Find the length of the tangent drawn from the point $(-5, 4)$ to the circle

   $$5x^2 + 5y^2 - 10x + 15y - 131 = 0$$

5. Find the length of the chord cut off from the line $2x+3y=13$ by the circle

$$x^2+y^2=26$$

6. Find the coordinates of the points of intersection of the line $x+2y=6$ with the circle: $x^2+y^2-2x-2y-39=0$

7. Find equations of the tangents to the circle $x^2+y^2=2$

(i) parallel to the $x-2y+1=0$

(ii) perpendicular to the line $3x+2y=6$

8. Find equations of the tangents drawn from

(i) $(0,5)$ to $x^2+y^2=16$

(ii) $(-1,2)$ to $x^2+y^2+4x+2y=0$

(iii) $(-7,-2)$ to $(x+1)^2+(y-2)^2=26$

Also find the points of contact.

9. Find an equation of the chord of contact of the tangents drawn from (4,5) to the circle

$$2x^2+2y^2-8x+12y+21=0$$

6.3 ANALYTIC PROOFS OF IMPORTANT PROPERTIES OF A CIRCLE

A line segment whose end points lie on a circle is called a **chord** of the circle. A **diameter** of a circle is a chord containing the centre of the circle.

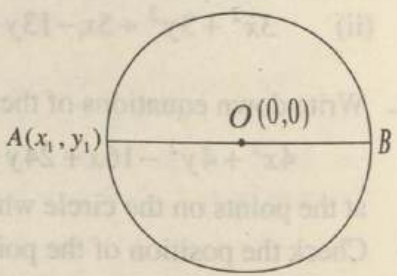
Theorem 1: Length of a diameter of the circle $x^2+y^2=a^2$ is $2a$.

Proof: Let AOB be a diameter of the circle

$$x^2+y^2=a^2 \qquad (1)$$

$O(0,0)$ is centre of (1).

Let the coordinates of A be (x_1, y_1).



Equation of AOB is

$$y=\frac{y_1}{x_1}x \qquad (2)$$

Substituting the value of y from (2) into (1), we have

$$x^2+\frac{y_1^2}{x_1^2}x^2=a^2 \quad \text{or} \quad x^2(x_1^2+y_1^2)=a^2x_1^2$$

Equation of PT is $y - y_1 = \frac{-x_1}{y_1}(x - x_1)$

or $yy_1 - y^2 = -xx_1 + x_1^2$

or $yy_1 + xx_1 = y_1^2 + x_1^2 = a^2$ ($\because P$ lies on the circle)

or $yy_1 + xx_1 - a^2 = 0$

Distance of PT from O (centre of the circle)

$$= \frac{|y_1(0) + x_1(0) - a^2|}{\sqrt{x^2 + y^2}} = \frac{|a^2|}{\sqrt{a^2}} = \frac{a^2}{a} = a \text{ (radius of the circle)}$$

Thus PT is tangent to the circle at $P(x_1, y_1)$

EXERCISE 6.3

1. Prove that normal lines of a circle pass through the centre of the circle.
2. Prove that the straight line drawn from the centre of a circle perpendicular to a tangent passes through the point of tangency.
3. Prove that the mid point of the hypotenuse of a right triangle is the circumcentre of the triangle.
4. Prove that the perpendicular dropped from a point of a circle on a diameter is a mean proportional between the segments into which it divides the diameter.

6.4 CONIC SECTION

In the following pages we shall study the remaining three conics.

Let L be a fixed line in a plane and F be a fixed point not on the line L. Suppose $|PM|$ denotes the distance of a point $P(x, y)$ from the line L. The set of all points P in the plane such that

$$\frac{|PF|}{|PM|} = e \quad \text{(a positive constant)}$$

is called a **conic section.**

(i) If $e = 1$, then the conic is a **parabola.**

(ii) If $0 < e < 1$, then the conic is an **ellipse.**

(iii) If $e > 1$, then the conic is a **hyperbola.**

EXERCISE 6.4

1. Find the focus, vertex and directix of the parabola sketch its graph.

(i) $y^2 = 8x$ (ii) $x^2 = -16y$ (iii) $x^2 = 5y$

(iv) $y^2 = -12x$ (v) $x^2 = 4(y-1)$ (vi) $y^2 = -8(x-3)$

(vii) $(x-1)^2 = 8(y+2)$ (viii) $y = 6x^2 - 1$ (ix) $x + 8 - y^2 + 2y = 0$

(x) $x^2 - 4x - 8y + 4 = 0$

2. Write an equation of the parabola with given elements.

(i) Focus $(-3, 1)$; directrix $x = 3$ (ii) Focus $(2, 5)$; directrix $y = 1$

(iii) Focus $(-3, 1)$; directrix $x - 2y - 3 = 0$ (iv) Focus $(1, 2)$; vertex $(3, 2)$

(v) Focus $(-1, 0)$; vertex $(-1, 2)$ (vi) Directrix $x = -2$, Focus $(2, 2)$

(vii) Directrix $y = 3$; vertex $(2, 2)$

(viii) Directrix $y = 1$, length of latusrectum is 8. Opens downward.

(ix) Axis $y = 0$, through $(2, 1)$ and $(11, -2)$

(x) Axis parallal to y-axis, the points $(0, 3)$, $(3, 4)$ and $(4, 11)$ lie on the graph.

3. Find an equation of the parabola having its focus at the origin and directrix parallel to the (i) the x-axis (ii) the y – axis.

4. Show that an equation of the parabola with focus at $(a\cos\alpha, a\sin\alpha)$ and directrix $x\cos\alpha + y\sin\alpha + a = 0$ is

$$(x\sin\alpha - y\cos\alpha)^2 = 4a(x\cos\alpha + y\sin\alpha)$$

Show that the ordinate at any point P of the parabola is a mean proportional between the length of the latusrectum and the abscissa of P.

6. A comet has a parabolic orbit with the earth at the focus. When the comet is 150,000 km from the earth, the line joining the comet and the earth makes an angle of 30 with the axis of the parabola. How close will the comet come to the earth?

7. Find an equation of the parabola formed by the cables of a suspension bridge whose span is a m and the vertical height of the supporting towers is b m.

8. A parabolic arch has a 100 m base and height 25 m. Find the height of the arch at the point 30 m from the centre of the base.

9. Show that tangent at any point P of a parabola makes equal angles with the line PF and the line through P parallel to the axis of the parabola, F being focus. (These angles are called respectively angle of **incidence** and angle of **reflection**).

6.5 ELLIPSE AND ITS ELEMENTS

We have already stated that a conic section is an ellipse if $e < 1$.

Let $0 < e < 1$ and F be a fixed point and L be a fixed line not containing F. Let $P(x, y)$ be a point in the plane and $|PM|$ be the perpendicular distance of P from L. The set of all points P such that

$$\frac{|PF|}{|PM|} = e$$

is called an ellipse.

The number e is **eccentricity** of the ellipse, F a **focus** and L a **directrix**.

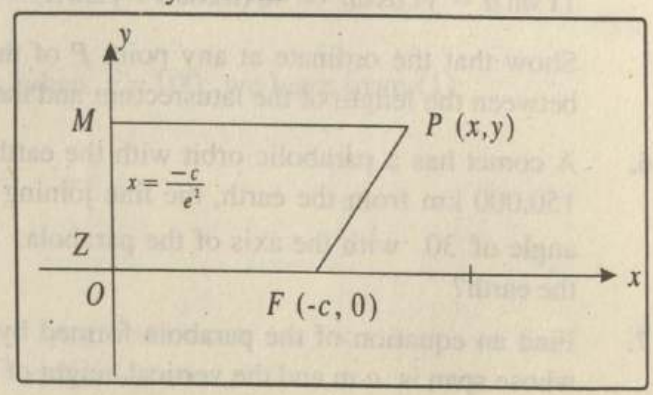
6.5.1 Standard Form of an Ellipse

Let $F(-c,0)$ be the focus and the line $x = \frac{-c}{e^2}$ be the directrix of an ellipse with eccentricity $e, (0 < e < 1)$. Let $P(x, y)$ be any point on the ellipse and suppose that $|PM|$ is the perpendicular distance of P from the directrix. Then

$$|PM| = x + \frac{c}{e^2}$$

The condition $|PF| = e|PM|$ takes the analytic form

$$(x+c)^2 + y^2 = e^2\left(x + \frac{c}{e^2}\right)^2$$



EXERCISE 6.5

1. Find an equation of the ellipse with given data and sketch its graph:

(i) Foci $(\pm 3,0)$ and minor axis of length 10

(ii) Foci $(0,-1)$ and $(0,-5)$ and major axis of length 6.

(iii) Foci $(-3\sqrt{3},0)$ and vertices $(\pm 6,0)$

(iv) Vertices $(-1,1)$, $(5,1)$; foci $(4,1)$ and $(0,1)$

(v) Foci $(\pm\sqrt{5},0)$ and passing through the point $\left(\frac{3}{2},\sqrt{3}\right)$

(vi) Vertices $(0,\pm 5)$, eccentricity $\frac{3}{5}$.

(vii) Centre $(0,0)$, focus $(0,-3)$, vertex $(0,4)$

(viii) Centre $(2,2)$, major axis parallel to y-axis and of length 8 units, minor axis parallel to x-axis and of length 6 units.

(ix) Centre $(0,0)$, symmetric with respect to both the axes and passing through the points $(2,3)$ and $(6,1)$.

(x) Centre $(0,0)$, major axis horizontal, the points $(3,1)$, $(4,0)$ lie on the graph.

2. Find the centre, foci, eccentricity, vertices and **directrices** of the ellipse **whose equation is given:**

(i) $x^2+4y^2=16$ (i) $9x^2+y^2=18$

(iii) $25x^2+9y^2=225$ (iv) $\frac{(2x-1)^2}{4}+\frac{(y+2)^2}{16}=1$

(v) $x^2+16x+4y^2-16y+76=0$ (vi) $25x^2+4y^2-250x-16y+541=0$

3. Let a be a positive number and $0<c<a$. Let $F(-c,0)$ and $F'(c,0)$ be two given points. Prove that the locus of points $P(x,y)$ such that

$|PF|+|PF'|=2a$, is an ellipse.

4. Use problem 3 to find equation of the ellipse as locus of points $P(x,y)$ such that the sum of the distances from P to the points $(0,0)$ and $(1,1)$ is 2.

5. Prove that the lactusrectum of the ellipse.

$$\frac{x^2}{a^2}+\frac{y^2}{b^2}=1 \text{ is } \frac{2b^2}{a}.$$

6. The major axis of an ellipse in standard form lies along the x-axis and has length $4\sqrt{2}$. The distance between the foci equals the length of the minor axis. Write an equation of the ellipse.
7. An astroid has elliptic orbit with the sun at one focus. Its distance from the sun ranges from 17 million miles to 183 million miles. Write an equation of the orbit of the astroid.
8. An arch in the shape of a semi-ellipse is 90m wide at the base and 30m high at the centre. At what distance from the centre is the arch $20\sqrt{2}$ m high?
9. The moon orbits the earth in an elliptic path with earth at one focus. The major and minor axes of the orbit are 768,806 km and 767,746 km respectively. Find the greatest and least distances (in Astronomy called the **apogee** and **perigee** respectively) of the moon from the earth.

6.6 HYPERBOLA AND ITS ELEMENTS

We have already stated that a conic section is a hyperbola if $e > 1$. Let $e > 1$ and F be a fixed point and L be a line not containing F. Also let $P(x, y)$ be a point in the plane and $|PM|$ be the perpendicular distance of P from L.

The set of all points $P(x, y)$ such that

$$\frac{|PF|}{|PM|} = e > 1 \qquad (1)$$

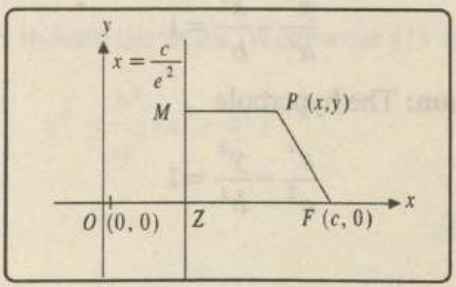
is called a **hyperbola**.

F and L are respectively **focus** and **directrix** of the hyperbola e is the eccentricity.

6.6.1 Standard Equation of Hyperbola

Let $F(c, 0)$ be the focus with $c > 0$ and $x = \frac{c}{e^2}$ be the directrix of the hyperbola. Also let $P(x, y)$ be a point on the hyperbola, then by definition

$$\frac{|PF|}{|PM|} = e$$



or $$\frac{(x-1)^2}{1^2}-\frac{(y+1)^2}{2^2}=1 \qquad (2)$$

We write $x-1=X,\ y+1=Y$ in (2), to have

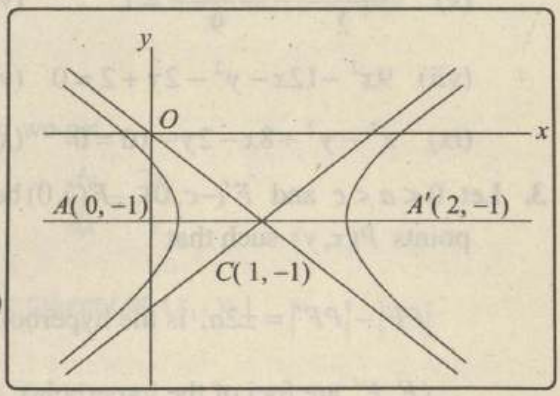
$$\frac{X^2}{1^2}-\frac{Y^2}{2^2}=1 \qquad (3)$$

so that it is a hyperbola with centre at $X=0,\ Y=0$ i.e., the centre of (1) is $(1, -1)$. The transverse axis of (3) is $Y=0$ i.e., $y+1=0$ is the transverse axis of (1). Vertices of (3) are: $X=\pm1,\ y=0$

i.e., $x-1=\pm1,\ y+1=0$ or $(0, -1)$ and $(2, -1)$

Here $a=1$ and $b=2$ so that, we have $c=\sqrt{c^2+b^2}=\sqrt{5}$

Eccentricity $e=\frac{c}{a}=\sqrt{5}$



Foci of (3) are : $X=\pm\sqrt{5},\ Y=0$

i.e., $x=1\pm\sqrt{5}$ and $y=-1$

i.e., $(1+\sqrt{5},-1)$ and $(1-\sqrt{5},-1)$

are foci of (1).

Equations of the directrices of (3) are: $X=\pm\frac{c}{e^2}=\pm\frac{\sqrt{5}}{5}=\pm\frac{1}{\sqrt{5}}$

or $x-1=\pm\frac{1}{\sqrt{5}}$ or $x=1+\frac{1}{\sqrt{5}}$ and $x=1-\frac{1}{\sqrt{5}}$

The sketch of the curve is as shown.

EXERCISE 6.6

1. Find an equation of the hyperbola with the given data. Sketch the graph of each.

(i) Centre (0, 0), focus (6, 0), vertex (4, 0)

(ii) Foci $(\pm5,0)$, vertex (3, 0)

(iii) Foci $(2\pm5\sqrt{2},-7)$, length of the transverse axis 10.

(iv) Foci $(0, \pm 6)$, $e=2$

(v) Foci $(0,\pm 9)$, directrices $y = \pm 4$.

(vi) Centre (2, 2), horizontal transverse axis of length 6 and eccentricity $e = 2$.

(vii) Vertices $(2, \pm 3)$, (0, 5) lies on the curve.

(viii) Foci (5, –2), (5, 4) and one vertex (5, 3).

2. Find the centre, foci, eccentricity, vertices and equations of diretrices of each of the following:

(i) $x^2 - y^2 = 9$ (ii) $\frac{x^2}{4} - \frac{y^2}{9} = 1$

(iii) $\frac{y^2}{16} - \frac{x^2}{9} = 1$ (iv) $\frac{y^2}{4} - x^2 = 1$

(v) $\frac{(x-1)^2}{2} - \frac{(y-1)^2}{9} = 1$ (vi) $\frac{(y+2)^2}{9} - \frac{(x-2)^2}{16} = 1$

(vii) $9x^2 - 12x - y^2 - 2y + 2 = 0$ (viii) $4y^2 + 12y - x^2 + 4x + 1 = 0$

(ix) $x^2 - y^2 + 8x - 2y - 10 = 0$ (x) $9x^2 - y^2 - 36x - 6y + 18 = 0$

3. Let $0 < a < c$ and $F'(-c, 0)$, $F(c, 0)$ be two fixed points. Show that the set of points $P(x, y)$ such that

$$|PF| - |PF'| = \pm 2a, \text{ is the hyperbola } \frac{x^2}{a^2} - \frac{y^2}{c^2 - a^2} = 1$$

(F, F' are foci of the hyperbola)

4. Using Problem 3, find an equation of the hyperbola with foci (–5, –5) and (5, 5), vertices $(-3\sqrt{2}, -3\sqrt{2})$ and $(3\sqrt{2}, 3\sqrt{2})$.

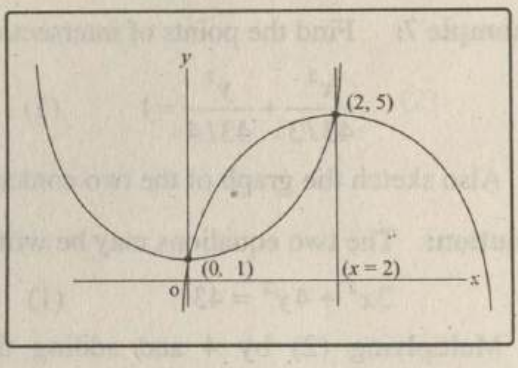
5. For any point on the hyperbola the difference of its distances from the points (2, 2) and (10, 2) is 6. Find an equation of the hyperbola.

6. Two listening posts hear the sound of an enemy gun. The difference in time is one second. If the listening posts are 1400 feet apart, write an equation of the hyperbola passing through the position of the enemy gum. (Sound travels at 1080 ft/sec).

6.7 TANGENTS AND NORMALS

We have already seen in the geometrical interpretation of the derivative of a curve $y = f(x)$ or $f(x, y) = 0$ that $\frac{dy}{dx}$ represents the slope of the tangent line to the

But $(-2, 5)$ does not satisfy (2). Thus $(0, 1)$ and $(2, 5)$ are the points of intersections of (1) and (2). $y = 1 + x^2$ is a parabola with vertex at $(0, 1)$ and opening upward. $y = 1 + 4x - x^2$ may be written as $y - 5 = -(x-2)^2$ which is a parabola with vertex $(2,5)$ and opening downward.



Example 9: Find equations of the common tangents to the two conics

$$\frac{x^2}{16} + \frac{y^2}{25} = 1 \quad \text{and} \quad \frac{x^2}{25} + \frac{y^2}{9} = 1$$

Solution: The tangents with slope m, to the two conics are respectively given by

$$y = mx \pm \sqrt{16m^2 + 25} \quad \text{and} \quad y = mx \pm \sqrt{25m^2 + 9}$$

For a tangent to be common, we must have

$$16m^2 + 25 = 25m^2 + 9$$

or $$9m^2 = 16 \quad \text{or} \quad m = \pm\frac{4}{3}$$

Using these values of m, equations of the four common tangents are:

$$y = \pm\frac{4}{3}x \pm \sqrt{481}.$$

EXERCISE 6.7

1. Find equations of the tangent and normal to each of the following at the indicated point:

   (i) $y^2 = 4ax$ at $(at^2, 2at)$

   (ii) $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ at $(a\cos\theta, b\sin\theta)$

   (iii) $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ at $(a\sec\theta, b\tan\theta)$

2. Write equation of the tangent to the given conic at the indicated point

   (i) $3x^2 = -16y$ at the points whose ordinate is -3.

(ii) $3x^2 - 7y^2 = 20$ at the points where $y = -1$.

(iii) $3x^2 - 7y^2 + 2x - y - 48 = 0$ at the point where $x = 4$.

3. Find equations of the tangents to each of the following through the given point:

(i) $x^2 + y^2 = 25$ through $(7, -1)$

(ii) $y^2 = 12x$ through $(1, 4)$

(iii) $x^2 - 2y^2 = 2$ through $(1, -2)$

4. Find equations of the normals to the parabola $y^2 = 8x$ which are parallel to the line $2x + 3y = 10$.

5. Find equations of the tangents to the ellipse $\frac{x^2}{4} + y^2 = 1$ which are parallel to the line $2x - 4y + 5 = 0$.

6. Find equations of the tangents to the conic $9x^2 - 4y^2 = 36$ parallel to $5x - 2y + 7 = 0$.

7. Find equations of the common tangents to the given conics

(i) $x^2 = 80y$ and $x^2 + y^2 = 81$

(ii) $y^2 = 16x$ and $x^2 = 2y$

8. Find the points of intersection of the given conics

(i) $\frac{x^2}{18} + \frac{y^2}{8} = 1$ and $\frac{x^2}{3} - \frac{y^2}{3} = 1$

(ii) $x^2 + y^2 = 8$ and $x^2 - y^2 = 1$

(iii) $3x^2 - 4y^2 = 12$ and $3y^2 - 2x^2 = 7$

(iv) $3x^2 + 5y^2 = 60$ and $9x^2 + y^2 = 124$

(v) $4x^2 + y^2 = 16$ and $x^2 + y^2 + y + 8 = 0$

$$5(X\cos\theta - Y\sin\theta)^2 + 2\sqrt{3}(X\cos\theta - Y\sin\theta)(X\sin\theta + Y\cos\theta)$$

$$+7(X\sin\theta + Y\cos\theta)^2 - 16 = 0 \qquad (1)$$

Since this equation is to be free from the product term XY, the coefficient of XY is zero, i.e., $-10\sin\theta\cos\theta + 2\sqrt{3}(\cos^2\theta - \sin^2\theta) + 14\sin\theta\cos\theta = 0$

or $\quad 2\sin 2\theta + 2\sqrt{3}\cos 2\theta = 0$

or $\quad \tan 2\theta = \dfrac{-2\sqrt{3}}{2} = \tan 120° \quad$ or $\quad \theta = 60°.$

Thus axes be rotated through an angle of $60°$ so that XY term is removed from the transformed equation.

Setting $\theta = 60°$ in (1), the transformed equation is (after simplification)

$$8X^2 + 4Y^2 - 16 = 0 \text{ or } 2X^2 + Y^2 - 4 = 0$$

EXERCISE 6.8

1. Find an equation of each of the following with respect to new parallel axes obtained by shifting the origin to the indicated point:

(i) $x^2 + 16y - 16 = 0, \quad O'(0, 1)$

(ii) $4x^2 + y^2 + 16x - 10y + 37 = 0, \quad O'(-2, 5)$

(iii) $9x^2 + 4y^2 + 18x - 16y - 11 = 0, \quad O'(-1, 2)$

iv) $x^2 - y^2 + 4x + 8y - 11 = 0, \quad O'(-2, 4)$

(v) $9x^2 - 4y^2 + 36x + 8y - 4 = 0, \quad O'(-2, 1)$

2. Find coordinates of the new origin (axes remaining parallel) so that first degree terms are removed from the transformed equation of each of the following. Also find the transformed equation:

(i) $3x^2 - 2y^2 + 24x + 12y + 24 = 0$ (ii) $25x^2 + 9y^2 + 50x - 36y - 164 = 0$

(iii) $x^2 - y^2 - 6x + 2y + 7 = 0$

3. In each of the following, find an equation referred to the new axes obtained by rotation of axes about the origin through the given angle:

(i) $xy = 1, \quad \theta = 45°$

(ii) $7x^2 - 8xy + y^2 - 9 = 0, \quad \theta = \arctan 2$

(iii) $9x^2+12xy+4y^2-x-y=0,\ \theta=\arctan\frac{2}{3}$

(iv) $x^2-2xy+y^2-2\sqrt{2}x-2\sqrt{2}y+2=0,\ \theta=45$

4. Find measure of the angle through which the axes be rotated so that the product term XY is removed from the transformed equation. Also find the transformed equation:

(i) $2x^2+6xy+10y^2-11=0$ (ii) $xy+4x-3y-10=0$

(ii) $5x^2-6xy+5y^2-8=0$

6.9 THE GENERAL EQUATION OF SECOND DEGREE

Standard equations of conic sections, namely circle parabola, ellipse and hyperbola have already been studied in the previous sections. Now we shall take up the general equation of second degree viz.

$$Ax^2+By^2+Gx+Fy+C=0 \qquad (1)$$

The nature of the curve represented by (1) can be determined by examining the coefficients A, B in the above equation. The following cases arise:

(i) If $A=B\neq 0$, equation (1) may be written as

$$A(x^2+y^2)+Gx+Fy+C=0 \text{ or } x^2+y^2+\frac{G}{A}x+\frac{F}{A}y+\frac{C}{A}=0$$

which represents a circle with centre at $\left(-\frac{G}{2A},-\frac{F}{2A}\right)$ and radius $\sqrt{\frac{G^2}{4A^2}+\frac{F^2}{4A^2}-\frac{C}{A}}$.

(ii) If $A\neq B$ and both are of the same sign, then we have

$$(Ax^2+Gx)+(By^2+Fy)+C=0$$

$$\text{or } A\left(x^2+\frac{G}{A}x+\frac{G^2}{4A^2}\right)+B\left(y^2+\frac{F}{B}y+\frac{F^2}{4B^2}\right)=\frac{G^2}{4A}+\frac{F^2}{4B}-C$$

$$\text{or } A\left(x+\frac{G}{2A}\right)^2+B\left(y+\frac{F}{2B}\right)^2=\frac{G^2}{4A}+\frac{F^2}{4B}-C \qquad (2)$$

If we write $X=x+\frac{G}{2A}$, $Y=y+\frac{F}{2B}$, then (2) can be written as

$$AX^2+BY^2=\frac{G^2}{4A}+\frac{F^2}{4B}-C=K \text{ (say) or } \frac{X^2}{(\sqrt{K/A})^2}+\frac{Y^2}{(\sqrt{K/B})^2}=1$$

which is standard equation of an **ellipse** in XY-coordinate system.

Example 6: **Use vectors, to prove that the diagonals of a parallelogram bisect each other.**

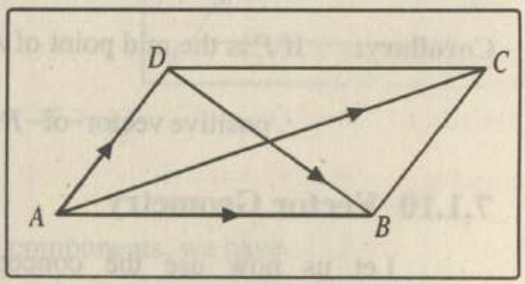
Solution: Let the vertices of the parallelogram be A, B, C and D (see figure)

Since $\overrightarrow{AC} = \overrightarrow{AB} + \overrightarrow{AD}$, the vector from A to the mid point of diagonal $\overrightarrow{AC}$ is

$$\underline{v} = \frac{1}{2}\left(\overrightarrow{AB} + \overrightarrow{AD}\right)$$

Since $\overrightarrow{DB} = \overrightarrow{AB} - \overrightarrow{AD}$, the vector from A to the mid point of diagonal $\overrightarrow{DB}$ is

$$\begin{aligned}\underline{w} &= \overrightarrow{AD} + \frac{1}{2}\left(\overrightarrow{AB} - \overrightarrow{AD}\right)\\ &= \overrightarrow{AD} + \frac{1}{2}\overrightarrow{AB} - \frac{1}{2}\overrightarrow{AD}\\ &= \frac{1}{2}\left(\overrightarrow{AB} + \overrightarrow{AD}\right)\\ &= \underline{v}\end{aligned}$$



Since $\underline{v} = \underline{w}$, these mid points of the diagonals $\overrightarrow{AC}$ and $\overrightarrow{DB}$ are the same.

Thus the diagonals of a parallelogram bisect each other.

EXERCISE 7.1

1. Write the vector $\overrightarrow{PQ}$ in the form $x\underline{i} + y\underline{j}$.

(i) $P = (2,3)$, $Q(6,-2)$ (ii) $P = (0,5)$, $Q(-1,-6)$

2. Find the magnitude of the vector $\underline{u}$:

(i) $\underline{u} = 2\underline{i} - 7\underline{j}$ (ii) $\underline{u} = \underline{i} + \underline{j}$ (iii) $\underline{u} = [3,-4]$

3. If $\underline{u} = 2\underline{i} - 7\underline{j}$, $\underline{v} = \underline{i} - 6\underline{j}$ and $\underline{w} = -\underline{i} + \underline{j}$. Find the following vectors:

(i) $\underline{u} + \underline{v} - \underline{w}$ (ii) $2\underline{u} - 3\underline{v} + 4\underline{w}$ (iii) $\frac{1}{2}\underline{u} + \frac{1}{2}\underline{v} + \frac{1}{2}\underline{w}$

4. Find the sum of the vectors $\overrightarrow{AB}$ and $\overrightarrow{CD}$, given the four points $A(1, -1)$, $B(2, 0)$, $C(-1, 3)$ and $D(-2, 2)$.

5. Find the vector from the point A to the origin where $\overrightarrow{AB} = 4\underline{i} - 2\underline{j}$ and B is the point $(-2, 5)$.

6. Find a unit vector in the direction of the vector given below:

(i) $\underline{v} = 2\underline{i} - \underline{j}$ (ii) $\underline{v} = \frac{1}{2}\underline{i} + \frac{\sqrt{3}}{2}\underline{j}$ (iii) $\underline{v} = -\frac{\sqrt{3}}{2}\underline{i} - \frac{1}{2}\underline{j}$

7. If A, B and C are respectively the points (2, -4), (4, 0) and (1, 6). Use vector method to find the coordinates of the point D if:

(i) $ABCD$ is a parallelogram (ii) $ADBC$ is a parallelogram

8. If B, C and D are respectively (4, 1), (–2, 3) and (–8, 0). Use vector method to find the coordinates of the point:

(i) A if $ABCD$ is a parallelogram. (ii) E if $AEBD$ is a parallelogram.

9. If O is the origin and $\overrightarrow{OP} = \overrightarrow{AB}$, find the point P when A and B are (–3, 7) and (1,0) respectively.

10. Use vectors, to show that $ABCD$ is a parallelogram, when the points A, B, C and D are respectively (0, 0), $(a, 0)$, (b, c) and $(b - a, c)$.

11. If $\overrightarrow{AB} = \overrightarrow{CD}$. Find the coordinates of the point A when points B, C, D are (1, 2), (–2, 5), (4, 11) respectively.

12. Find the position vectors of the point of division of the line segments joining the following pair of points, in the given ratio:

(i) Point C with position vector $2\underline{i} - 3\underline{j}$ and point D with position vector $3\underline{i} + 2\underline{j}$ in the ratio 4 : 3

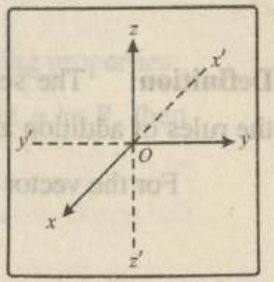
(ii) Point E with position vector $5\underline{i}$ and point F with position vector $4\underline{i} + \underline{j}$ in ratio 2 : 5

14. Prove that the line segment joining the mid points of two sides of a triangle is parallel to the third side and half as long.

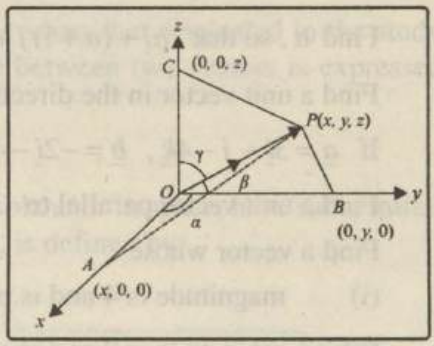
15. Prove that the line segments joining the mid points of the sides of a quadrilateral taken in order form a parallelogram.

7.2 INTRODUCTION OF VECTOR IN SPACE

In space, a rectangular coordinate system is constructed using three mutually orthogonal (perpendicular) axes, which have origin as their common point of intersection. When sketching figures, we follow the convention that the positive x-axis points towards the reader, the positive y-axis to the right and the positive z-axis points upwards.



(ii) the numbers $\cos\alpha$, $\cos\beta$ and $\cos\gamma$ are called direction cosines of the vector $\underline{r}$.



Important Result:

Prove that $\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$

Proof:

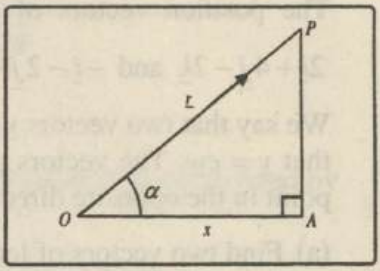
Let $\underline{r} = [x, y, z] = x\underline{i} + y\underline{j} + z\underline{k}$

$\therefore \quad |\underline{r}| = \sqrt{x^2 + y^2 + z^2} = r$

then $\dfrac{\underline{r}}{|\underline{r}|} = \left[\dfrac{x}{r}, \dfrac{y}{r}, \dfrac{z}{r}\right]$ is the unit vector in the direction of the vector $\underline{r} = \overrightarrow{OP}$.

It can be visualized that the triangle OAP is a right triangle with $\angle A = 90$.

Therefore in right triangle OAP,



$$\cos\alpha = \frac{\overline{OA}}{\overline{OP}} = \frac{x}{r}, \text{ similarly}$$

$$\cos\beta = \frac{y}{r}, \quad \cos\gamma = \frac{z}{r}$$

The numbers $\cos\alpha = \dfrac{x}{r}$, $\cos\beta = \dfrac{y}{r}$ and $\cos\gamma = \dfrac{z}{r}$ are called the direction cosines of $\overrightarrow{OP}$.

$$\therefore \quad \cos^2\alpha + \cos^2\beta + \cos^2\gamma = \frac{x^2}{r^2} + \frac{y^2}{r^2} + \frac{z^2}{r^2} = \frac{x^2 + y^2 + z^2}{r^2} = \frac{r^2}{r^2} = 1$$

EXERCISE 7.2

1. Let $A = (2, 5)$, $B = (-1, 1)$ and $C = (2, -6)$. Find

(i) $\overrightarrow{AB}$ (ii) $2\overrightarrow{AB} - \overrightarrow{CB}$ (iii) $2\overrightarrow{CB} - 2\overrightarrow{CA}$

2. Let $\underline{u} = \underline{i} + 2\underline{j} - \underline{k}$, $\underline{v} = 3\underline{i} - 2\underline{j} + 2\underline{k}$, $\underline{w} = 5\underline{i} - \underline{j} + 3\underline{k}$. Find the indicated vector or number.

(i) $\underline{u} + 2\underline{v} + \underline{w}$ (ii) $\underline{v} - 3\underline{w}$ (iii) $|3\underline{v} + \underline{w}|$

3. Find the magnitude of the vector $\underline{v}$ and write the direction cosines of $\underline{v}$.
(i) $\underline{v} = 2\underline{i} + 3\underline{j} + 4\underline{k}$ (ii) $\underline{v} = \underline{i} - \underline{j} - \underline{k}$ (iii) $\underline{v} = 4\underline{i} - 5\underline{j}$.

4. Find α, so that $\left|\alpha\underline{i} + (\alpha+1)\underline{j} + 2\underline{k}\right| = 3$.

5. Find a unit vector in the direction of $\underline{v} = \underline{i} + 2\underline{j} - \underline{k}$.

6. If $\underline{a} = 3\underline{i} - \underline{j} - 4\underline{k}$, $\underline{b} = -2\underline{i} - 4\underline{j} - 3\underline{k}$ and $\underline{c} = \underline{i} + 2\underline{j} - \underline{k}$.
Find a unit vector parallel to $3\underline{a} - 2\underline{b} + 4\underline{c}$.

7. Find a vector whose
(i) magnitude is 4 and is parallel to $2\underline{i} - 3\underline{j} + 6\underline{k}$
(ii) magnitude is 2 and is parallel to $-\underline{i} + \underline{j} + \underline{k}$.

8. If $\underline{u} = 2\underline{i} + 3\underline{j} + 4\underline{k}$, $\underline{v} = -\underline{i} + 3\underline{j} - \underline{k}$ and $\underline{w} = \underline{i} + 6\underline{j} + z\underline{k}$ represent the sides of a triangle. Find the value of z.

9. The position vectors of the points A, B, C and D are $2\underline{i} - \underline{j} + \underline{k}$, $3\underline{i} + \underline{j}$, $2\underline{i} + 4\underline{j} - 2\underline{k}$ and $-\underline{i} - 2\underline{j} + \underline{k}$ respectively. Show that $\overrightarrow{AB}$ is parallel to $\overrightarrow{CD}$.

10. We say that two vectors $\underline{v}$ and $\underline{w}$ in space are parallel if there is a scalar c such that $\underline{v} = c\underline{w}$. The vectors point in the same direction if $c > 0$ and the vectors point in the opposite direction if $c < 0$
(a) Find two vectors of length 2 parallel to the vector $\underline{v} = 2\underline{i} - 4\underline{j} + 4\underline{k}$.
(b) Find the constant a so that the vectors $\underline{v} = \underline{i} - 3\underline{j} + 4\underline{k}$ and $\underline{w} = a\underline{i} + 9\underline{j} - 12\underline{k}$ are parallel.
(c) Find a vector of length 5 in the direction opposite that of $\underline{v} = \underline{i} - 2\underline{j} + 3\underline{k}$.
(d) Find a and b so that the vectors $3\underline{i} - \underline{j} + 4\underline{k}$ and $a\underline{i} + b\underline{j} - 2\underline{k}$ are parallel.

11. Find the direction cosines for the given vector:
(i) $\underline{v} = 3\underline{i} - \underline{j} + 2\underline{k}$ (ii) $6\underline{i} - 2\underline{j} + \underline{k}$
(iii) $\overrightarrow{PQ}$, where $P = (2, 1, 5)$ and $Q = (1, 3, 1)$.

12. Which of the following triples can be the direction angles of a single vector:
(i) $45°, 45°, 60°$ (ii) $30°, 45°, 60°$ (iii) $45°, 60°, 60°$.

$\Rightarrow \quad = \underline{b}.\underline{b} + \underline{b}.\underline{c} + \underline{c}.\underline{b} + \underline{c}.\underline{c}$

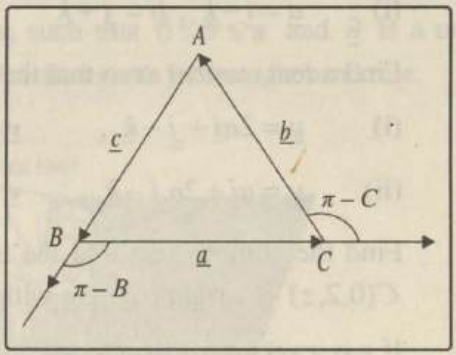
$\Rightarrow \quad a^2 = b^2 + 2\underline{b}.\underline{c} + c^2 \qquad (\because \ \underline{b}.\underline{c} = \underline{c}.\underline{b})$

$\Rightarrow \quad a^2 = b^2 + c^2 + 2bc.cos(\pi - A)$

$\therefore \quad a^2 = b^2 + c^2 - 2bc\cos A$

(ii) $\quad \underline{a} + \underline{b} + \underline{c} = \underline{0}$

$\Rightarrow \quad \underline{a} = -\underline{b} - \underline{c}$



Take dot product with $\underline{a}$

$\underline{a}.\underline{a} = -\underline{a}.\underline{b} - \underline{a}.\underline{c}$

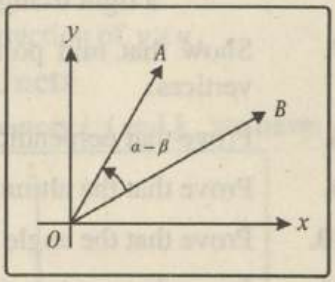
$= -ab\cos(\pi - C) - ac\cos(\pi - B)$

$a^2 = ab\cos C + ac\cos B$

$\Rightarrow a = b\cos C + c\cos B$

Example 9: Prove that: $cos(\alpha - \beta) = cos\alpha\, cos\beta + sin\alpha\, sin\beta$

Solution: Let $\overrightarrow{OA}$ and $\overrightarrow{OB}$ be the unit vectors in the xy-plane making angles α and β with the positive x-axis.



So that $\angle AOB = \alpha - \beta$

Now $\overrightarrow{OA} = cos\alpha\,\underline{i} + sin\alpha\,\underline{j}$

and $\overrightarrow{OB} = cos\beta\,\underline{i} + sin\beta\,\underline{i}$

$\therefore \ \overrightarrow{OA}.\overrightarrow{OB} = (cos\alpha\,\underline{i} + sin\alpha\,\underline{j}).\,(cos\beta\,\underline{i} + sin\beta\,\underline{j})$

$\Rightarrow \ \left|\overrightarrow{OA}\right|\left|\overrightarrow{OB}\right| \ cos(\alpha - \beta) = cos\alpha\, cos\beta + sin\alpha\, sin\beta$

$\therefore \ cos(\alpha - \beta) = cos\alpha\, cos\beta + sin\alpha\, sin\beta \qquad \left(\because \left|\overrightarrow{OA}\right| = \left|\overrightarrow{OB}\right| = 1\right)$

EXERCISE 7.3

1. Find the cosine of the angle θ between $\underline{u}$ and $\underline{v}$:

(i) $\quad \underline{u} = 3\underline{i} + \underline{j} - \underline{k}$, $\underline{v} = 2\underline{i} - \underline{j} + \underline{k}$ (ii) $\quad \underline{u} = \underline{i} - 3\underline{j} + 4\underline{k}$, $\underline{v} = 4\underline{i} - \underline{j} + 3\underline{k}$

(iii) $\underline{u}=[-3, 5]$, $\underline{v}=[6, -2]$ (iv) $\underline{u}=[2,-3, 1]$, $\underline{v}=[2, 4, 1]$

2. Calculate the projection of $\underline{a}$ along $\underline{b}$ and projection of $\underline{b}$ along $\underline{a}$ when:

(i) $\underline{a}=\underline{i}-\underline{k}$, $\underline{b}=\underline{j}+\underline{k}$ (ii) $\underline{a}=3\underline{i}+\underline{j}-\underline{k}$, $\underline{b}=-2\underline{i}-\underline{j}+\underline{k}$

3. Find a real number α so that the vectors $\underline{u}$ and $\underline{v}$ are perpendicular.

(i) $\underline{u}=2\alpha\underline{i}+\underline{j}-\underline{k}$, $\underline{v}=\underline{i}+\alpha\underline{j}+4\underline{k}$

(ii) $\underline{u}=\alpha\underline{i}+2\alpha\underline{j}-\underline{k}$, $\underline{v}=\underline{i}+\alpha\underline{j}+3\underline{k}$

4. Find the number z so that the triangle with vertices $A(1,-1,0)$, $B(-2,2,1)$ and $C(0,2,z)$ is a right triangle with right angle at C.

5. If $\underline{v}$ is a vector for which

$\underline{v}.\underline{i}=0$, $\underline{v}.\underline{j}=0$, $\underline{v}.\underline{k}=0$, find $\underline{v}$.

6. (i) Show that the vectors $3\underline{i}-2\underline{j}+\underline{k}$, $\underline{i}-3\underline{j}+5\underline{k}$ and $2\underline{i}+\underline{j}-4\underline{k}$ form a right angle.

(ii) Show that the set of points $P=(1,3,2)$, $Q(4,1,4)$ and $R=(6,5,5)$ form a right triangle.

7. Show that mid point of hypotenuse a right triangle is equidistant from its vertices.

8. Prove that perpendicular bisectors of the sides of a triangle are concurrent.

9. Prove that the altitudes of a triangle are concurrent.

10. Prove that the angle in a semi circle is a right angle.

11. Prove that $\cos(\alpha+\beta)=\cos\alpha\cos\beta-\sin\alpha\sin\beta$

12. Prove that in any triangle ABC.

(i) $b=c\cos A+a\cos C$ (ii) $c=a\cos B+b\cos A$

(iii) $b^2=c^2+a^2-2ca\cos B$ (iv) $c^2=a^2+b^2-2ab\cos C$.

7.4 THE CROSS PRODUCT OR VECTOR PRODUCT OF TWO VECTORS

The vector product of two vectors is widely used in Physics, particularly Mechanics and Electricity. It is only defined for vector in space.

Now $\overrightarrow{PQ}\times\overrightarrow{PR}=\begin{vmatrix}\underline{i} & \underline{j} & \underline{k}\\ -1 & 2 & 4\\ 2 & -1 & 4\end{vmatrix}=(8+4)\underline{i}-(-4-8)\underline{j}+(1-4)\underline{k}$

$\therefore$ Area of parallelogram $=\left|\overrightarrow{PQ}\times\overrightarrow{PR}\right|=\left|12\underline{i}+12\underline{j}-3\underline{k}\right|$

$=\sqrt{144+144+9}$

$=\sqrt{297}$

Be careful!:
Not all pairs of vertices give a side e.g. $\overrightarrow{PS}$ is not a side, it is diagonal since $\overrightarrow{PQ}+\overrightarrow{PR}=\overrightarrow{PS}$

Example7: If $\underline{u}=2\underline{i}-\underline{j}+\underline{k}$ and $\underline{v}=4\underline{i}+2\underline{j}-\underline{k}$, find by determinant formula

(i) $\underline{u}\times\underline{u}$ (ii) $\underline{u}\times\underline{v}$ (iii) $\underline{v}\times\underline{u}$

Solution: $\underline{u}=2\underline{i}-\underline{j}+\underline{k}$ and $\underline{v}=4\underline{i}+2\underline{j}-\underline{k}$

By determinant formula

(i) $\underline{u}\times\underline{u}=\begin{vmatrix}\underline{i} & \underline{j} & \underline{k}\\ 2 & -1 & 1\\ 2 & -1 & 1\end{vmatrix}=0$ ($\therefore$ Two rows are same)

(ii) $\underline{u}\times\underline{v}=\begin{vmatrix}\underline{i} & \underline{j} & \underline{k}\\ 2 & -1 & 1\\ 4 & 2 & -1\end{vmatrix}=(1-2)\underline{i}-(-2-4)\underline{j}+(4+4)\underline{k}=-\underline{i}+6\underline{j}+8\underline{k}$

(iii) $\underline{v}\times\underline{u}=\begin{vmatrix}\underline{i} & \underline{j} & \underline{k}\\ 4 & 2 & -1\\ 2 & -1 & 1\end{vmatrix}=(2-1)\underline{i}-(4+2)\underline{j}+(-4-4)\underline{k}=\underline{i}-6\underline{j}-8\underline{k}$

EXERCISE 7.4

1. Compute the cross product $\underline{a}\times\underline{b}$ and $\underline{b}\times\underline{a}$. Check your answer by showing that each $\underline{a}$ and $\underline{b}$ is perpendicular to $\underline{a}\times\underline{b}$ and $\underline{b}\times\underline{a}$.

(i) $\underline{a}=2\underline{i}+\underline{j}-\underline{k}$, $\underline{b}=\underline{i}-\underline{j}+\underline{k}$ (ii) $\underline{a}=\underline{i}+\underline{j}$, $\underline{b}=\underline{i}-\underline{j}$

(iii) $\underline{a}=3\underline{i}-2\underline{j}+\underline{k}$, $\underline{b}=\underline{i}+\underline{j}$ (iv) $\underline{a}=-4\underline{i}+\underline{j}-2\underline{k}$, $\underline{b}=2\underline{i}+\underline{j}+\underline{k}$

2. Find a unit vector perpendicular to the plane containing $\underline{a}$ and $\underline{b}$. Also find sine of the angle between them.

(i) $\underline{a}=2\underline{i}-6\underline{j}-3\underline{k}$, $\underline{b}=4\underline{i}+3\underline{j}-\underline{k}$ (ii) $\underline{a}=-\underline{i}-\underline{j}-\underline{k}$, $\underline{b}=2\underline{i}-3\underline{j}+4\underline{k}$

(iii) $\underline{a}=2\underline{i}-2\underline{j}+4\underline{k}$, $\underline{b}=-\underline{i}+\underline{j}-2\underline{k}$ (iv) $\underline{a}=\underline{i}+\underline{j}$, $\underline{b}=\underline{i}-\underline{j}$

3. Find the area of the triangle, determined by the point P, Q and R.

(i). $P(0, 0, 0)$; $Q(2, 3, 2)$; $R(-1, 1, 4)$ (ii) $P(1, -1, -1)$; $Q(2, 0, -1)$; $R(0, 2, 1)$

4. Find the area of parallelogram, whose vertices are:

(i) $A(0, 0, 0)$; $B(1, 2, 3)$; $C(2, -1, 1)$; $D(3, 1, 4)$

(ii) $A(1, 2, -1)$; $B(4, 2, -3)$; $C(6, -5, 2)$; $D(9, -5, 0)$

(iii) $A(-1, 1, 1)$; $B(-1, 2, 2)$; $C(-3, 4, -5)$; $D(-3, 5, -4)$

5. Which vectors, if any, are perpendicular or parallel

(i) $\underline{u}=5\underline{i}-\underline{j}+\underline{k}$; $\underline{v}=\underline{j}-5\underline{k}$; $\underline{w}=-15\underline{i}+3\underline{j}-3\underline{k}$

(ii) $\underline{u}=\underline{i}+2\underline{j}-\underline{k}$; $\underline{v}=-\underline{i}+\underline{j}+\underline{k}$; $\underline{w}=-\frac{\pi}{2}\underline{i}-\pi\underline{j}+\frac{\pi}{2}\underline{k}$

6. Prove that: $\underline{a}\times(\underline{b}+\underline{c})+\underline{b}\times(\underline{c}+\underline{a})+\underline{c}\times(\underline{a}+\underline{b})=0$

7. If $\underline{a}+\underline{b}+\underline{c}=0$, then prove that $\underline{a}\times\underline{b}=\underline{b}\times\underline{c}=\underline{c}\times\underline{a}$

8. Prove that: $\sin(\alpha-\beta)=\sin\alpha\cos\beta+\cos\alpha\sin\beta$.

9. If $\underline{a}\times\underline{b}=0$ and $\underline{a}\cdot\underline{b}=0$, what conclusion can be drawn about $\underline{a}$ or $\underline{b}$?

7.5 SCALAR TRIPLE PRODUCT OF VECTORS

There are two types of triple product of vectors:

(a) **Scalar Triple Product:** $(\underline{u}\times\underline{v})\cdot\underline{w}$ or $\underline{u}\cdot(\underline{v}\times\underline{w})$

(b) **Vector Triple product:** $\underline{u}\times(\underline{v}\times\underline{w})$

In this section we shall study the scalar triple product only

Definition

Let $\underline{u}=a_1\underline{i}+b_1\underline{j}+c_1\underline{k}$, $\underline{v}=a_2\underline{i}+b_2\underline{j}+c_2\underline{k}$ and $\underline{w}=a_3\underline{i}+b_3\underline{j}+c_3\underline{k}$ be three vectors

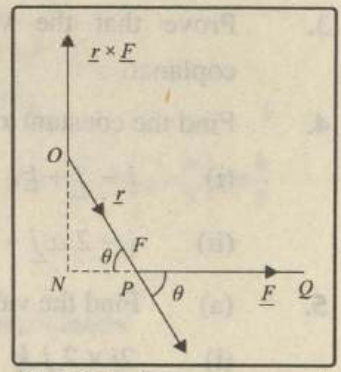
The scalar triple product of vector $\underline{u}$, $\underline{v}$ and $\underline{w}$ is defined by

$\underline{u}.(\underline{v}\times\underline{w})$ or $\underline{v}.(\underline{w}\times\underline{u})$ or $\underline{w}.(\underline{u}\times\underline{v})$

The scalar triple product $\underline{u}.(\underline{v}\times\underline{w})$ is written as

(b) Moment of Force

Let a force $\underline{F}$ $(\overrightarrow{PQ})$ act at a point P as shown in the figure, then



moment of $\underline{F}$ about O.

= product of force $\underline{F}$ and perpendicular ON . $\hat{\underline{n}}$

$= (PQ)(ON)(\hat{\underline{n}}) = (PQ)(OP)\sin\theta\,.\,\hat{\underline{n}}$

$= \overrightarrow{OP} \times \overrightarrow{PQ} = \underline{r} \times \underline{F}$

Example 8: Find the moment about the point $M(-2,4,-6)$ of the force represented by $\overrightarrow{AB}$, where coordinates of points A and B are $(1, 2, -3)$ and $(3, -4, 2)$ respectively.

Solution: $\overrightarrow{AB} = (3-1)\underline{i} + (-4-2)\underline{j} + (2+3)\underline{k} = 2\underline{i} - 6\underline{j} + 5\underline{k}$

$\overrightarrow{MA} = (1+2)\underline{i} + (2-4)\underline{j} + (-3+6)\underline{k} = 3\underline{i} - 2\underline{j} + 3\underline{k}$

Moment of $\overrightarrow{AB}$ about $(-2, 4, -6) = \underline{r} \times \underline{F} = \overrightarrow{MA} \times \overrightarrow{AB}$

$$= \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ 3 & -2 & 3 \\ 2 & -6 & 5 \end{vmatrix}$$

$= (-10+18)\underline{i} - (15-6)\underline{j} + (-18+4)\underline{k}$

$= 8\underline{i} - 9\underline{j} - 14\underline{k}$

Magnitude of the moment $= \sqrt{(8)^2 + (-9)^2 + (-14)^2} = \sqrt{341}$

EXERCISE 7.5

1. Find the volume of the parallelepiped for which the given vectors are three edges.

(i) $\underline{u} = 3\underline{i} + 2\underline{k}$; $\underline{v} = \underline{i} + 2\underline{j} + \underline{k}$; $\underline{w} = -\underline{j} + 4\underline{k}$

(ii) $\underline{u} = \underline{i} - 4\underline{j} - \underline{k}$; $\underline{v} = \underline{i} - \underline{j} - 2\underline{k}$; $\underline{w} = 2\underline{i} - 3\underline{j} + \underline{k}$

(iii) $\underline{u} = \underline{i} - 2\underline{j} + 3\underline{k}$; $\underline{v} = 2\underline{i} - \underline{j} - \underline{k}$; $\underline{w} = \underline{j} + \underline{k}$

2. Verify that

$\underline{a}, \underline{b} \times \underline{c} = \underline{b}, \underline{c} \times \underline{a} = \underline{c}, \underline{a} \times \underline{b}$

if $\underline{a} = 3\underline{i} - \underline{j} + 5k$, $\underline{b} = 4\underline{i} + 3\underline{j} - 2\underline{k}$ and $\underline{c} = 2\underline{i} + 5\underline{j} + \underline{k}$

3. Prove that the vectors $\underline{i}-2\underline{j}+3\underline{k}$, $-2\underline{i}+3\underline{j}-4\underline{k}$ and $\underline{i}-3\underline{j}+5\underline{k}$ are coplanar.

4. Find the constant α such that the vectors are coplanar.

   (i) $\underline{i}-\underline{j}+\underline{k}$, $\underline{i}-2\underline{j}-3\underline{k}$ and $3\underline{i}-\alpha\underline{j}+5\underline{k}$,

   (ii) $\underline{i}-2\alpha\underline{j}-\underline{k}$, $\underline{i}-\underline{j}+2\underline{k}$ and $\alpha\underline{i}-\underline{j}+\underline{k}$

5. (a) Find the value of:

   (i) $2\underline{i}\times 2\underline{j}.\underline{k}$ (ii) $3\underline{j}.\underline{k}\times\underline{i}$ (iii) $[\underline{k}\ \underline{i}\ \underline{j}]$ (iv) $[\underline{i}\ \underline{i}\ \underline{k}]$

   (b) Prove that $\underline{u}.(\underline{v}\times\underline{w})+\underline{v}.(\underline{w}\times\underline{u})+\underline{w}.(\underline{u}\times\underline{v})=3\underline{u}.(\underline{v}\times\underline{w})$

6. Find volume of the Tetrahedron with the Vertices

   (i) (0, 1, 2), (3, 2, 1), (1, 21) and (5, 5, 6)

   (ii) (2, 1, 8), (3, 2, 9), (2, 1, 4) and (3, 3, 10).

7. Find the work done, if the point at which the constant force $\underline{F}=4\underline{i}+3\underline{j}+5\underline{k}$ is applied to an object, moves from $P_1(3,1,-2)$ to $P_2(2,4,6)$.

8. A particle, acted by constant forces $4\underline{i}+\underline{j}-3\underline{k}$ and $3\underline{i}-\underline{j}-\underline{k}$, is displacement from $A(1,2,3)$ to $B(5,4,1)$. Find the word done.

9. A particle is displaced from the point $A(5,-5,-7)$ to the point $B(6,2,-2)$ under the action of constant forces defined by $10\underline{i}-\underline{j}+11\underline{k}$, $4\underline{i}+5\underline{j}+9\underline{k}$ and $-2\underline{i}+\underline{j}-9\underline{k}$. Show that the total work done by the forces is 102 units.

10. A force of magnitude 6 units acting parallel to $2\underline{i}-2\underline{j}+\underline{k}$ displaces, the point of application from (1, 2,3) to (5,3,7). Find the work done.

11. A force $\underline{F}=3\underline{i}+2\underline{j}-4\underline{k}$ is applied at the point (1,–1, 2). Find the moment of the force about the point (2, –1, 3).

12. A force $\underline{F}=4\underline{i}-3\underline{k}$, passes through the point $A(2,-2,5)$. Find the moment of $\underline{F}$ about the point $B(1,-3,1)$.

13. Give a force $\underline{F}=2\underline{i}+\underline{j}-3\underline{k}$ acting at a point $A(1,-2,1)$. Find the moment of $\underline{F}$ about the point $B(2,0,-2)$.

14. Find the moment about $A(1,1,1)$ of each of the concurrent forces $\underline{i}-2\underline{j}$, $3\underline{i}+2\underline{j}-\underline{k}$, $5\underline{j}+2\underline{k}$, where $P(2,0,1)$ is their point of concurrency.

15. A force $F=7\underline{i}+4\underline{j}-3\underline{k}$ is applied at $P(1,-2,3)$. Find its moment about the point $Q(2,1,1)$.